

Foreign Aid Between Disaster and Rebellion: Subnational Evidence from the Philippines

Abstract: How do states balance between allocating foreign aid to manage disasters or to manage internal conflict? This paper argues that the saliency of both conflict and natural disasters influences where foreign aid projects are prioritized. Specifically, states prioritize disaster-impacted localities where government control of said localities is high to ensure that aid is not co-opted or destroyed by insurgents and to maintain or improve stability. This paper evaluates this theory through a subnational analysis of the Philippines' municipalities from 2011 to 2014. World Bank project aid is more prevalent in government-controlled territories that are impacted by a natural disaster than in similarly affected rebel-controlled territories. Contested municipalities are the most likely and most funded places for World Bank aid. These findings contribute to important questions as to the role of recipient states in aid delivery and whether foreign aid is driven by strategic needs rather than humanitarian needs.

Introduction

How do states balance between utilizing foreign aid to manage disasters or to manage conflict? Foreign aid can be a crucial lifeline for many communities during times of war. Before the collapse of the Afghan government in August 2021, for example, Afghanistan was immensely reliant on foreign aid, financing over half of the government's \$6 billion annual budget before its collapse (Runde et al., 2024) and making up 40% of the nation's GDP (Runde et al., 2024; Cordesman, 2022). In 2020, over \$700 million in foreign aid was recorded by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) Financial Tracking Services, provided from a variety of donors, such as the U.S., U.K., and Japan, as well as international organizations such as the European Commission or World Bank (Afghanistan 2020 | Financial Tracking Service, n.d.).

Much of this aid was in response to challenges only exacerbated by the conflict. Over 25% of this aid went towards food security or nutrition-related projects, with other primary areas of funding being health or water and sanitation-focused (over 10% of aid) or protection-focused (over 7%) (Afghanistan 2020 | Financial Tracking Service, n.d.). Afghanistan also faces challenges from natural disasters. Afghanistan's conflict weakened its ability to manage natural disasters, causing a high humanitarian cost that places Afghanistan first in the world for the impact of disasters on its population (Mena & Hilhorst, 2021). Afghanistan is not the only state that faces such balances between disasters and conflict, however. Natural disasters coincide with conflict in a variety of cases globally – Syria, Nigeria, Somalia, Kashmir, the Philippines, among others, have all had natural disasters impact them during conflict (Ide, 2023c). States facing conflict face unique pressures and priorities, creating trade-offs that may not be present in non-

conflict contexts. Given the immense toll both natural disasters and conflict take on the state and its people, how do governments determine where to prioritize the use of foreign aid?

I argue that, when states face disasters in the midst of conflict, states seek to balance humanitarian needs and strategic incentives by providing foreign aid to disaster-affected regions in areas and times where the government has territorial control. Aid has strategic implications while also serving as a tool for enhancing state stability, incentivizing the state to make considerations as to the level of control it has over a given locality before dispensing aid. Because of this, territorial control may impact the relationship between natural disasters and foreign aid. I further argue that while aid to rebel territory provides strategic incentives for the government, the risk of co-optation or destruction makes it less desirable than government-controlled territory, but still a viable option compared to contested territories, where the likelihood of attribution of success is low and the risk of destruction is high.

This paper explores the allocation of foreign aid at the local level. Specifically, I examine aid distribution from the World Bank at a municipality-level in the Philippines, from 2011 to 2014. I use geocoded data on foreign aid, conflict, and natural disasters at the municipality-month level to examine my theory. Through examining a singular case of conflict and focusing on the subnational dispersion of foreign aid within the state, this paper provides greater insights into the dynamic between natural disasters, aid, and conflict by being able to better discern *where* foreign aid is going once it has been decided that a recipient state will receive it.

This paper identifies and reconciles a puzzle revealed by previous research at the intersection of foreign aid, conflict, and natural disasters (Desportes, 2019; Desportes and Hilhorst, 2020; Mena and Hilhorst, 2021). Each literature provides internally consistent findings. Literature on foreign aid and conflict suggests that aid may incentivize conflict due to its utility as a resource

and its potential fungibility (Findley et al., 2011; Narang, 2015; Wood & Molfino, 2016). The literature on disasters and foreign aid indicates that natural disasters are often found to increase the amount of foreign aid in the aftermath, which could provide resources or other benefits to the state and rebels (Mogge et al., 2023; Becerra et al., 2014; Francken et al., 2012; Strömberg, 2007).

However, the literature on the dynamics of natural disasters and civil conflict often argues that disasters can deprive impacted rebels and governments of resources, increasing the duration of conflict (Eastin, 2016; Keels, 2019; Ide, 2023b; Ide, 2023a). If natural disasters lead to increased aid to recipient states, but if impacted parties often still feel the impacts of resource deprivation, is aid not being provided in conflict zones that face disasters, and rather prioritized to non-conflict environments within the state? Or is aid being provided in conflict zones, with the effects of the disaster persisting even with aid provision? By conducting a subnational analysis of municipalities within the Philippines, this analysis examines where aid projects are going in relation to where disasters and conflict are impacting localities, providing more direct evidence about the interdependencies between these three factors to answer these questions.

I find that local aid project placement and funding in disaster-affected municipalities are moderated by the level of territorial control the government. As territorial control shifts from the government to the rebels, contested territories receive more aid than either fully government or rebel-controlled municipalities, but government-controlled municipalities (e.g., where 0% of the municipality is controlled by the rebels) are still more likely to receive aid and receive more funding than rebel-controlled municipalities (e.g., where 100% of the municipality is controlled by the rebels). These findings highlight that while international donors have some influence over

aid placement, the interests of recipient states may still shape aid placement rather than solely the humanitarian need of those within the state, supporting previous research (Rosvold, 2020).

Literature Review

Foreign Aid and Conflict

When looking at the broad effects of foreign aid in civil conflict, a rich body of research examines foreign aid's impact on conflict processes such as duration, onset, and intensity. Aid provision has often been found to increase violence (Findley et al., 2011; Sexton, 2016; Wood & Molfino, 2016). For instance, the presence of aid encourages rebel looting (Wood & Sullivan, 2015; Addison et al., 2002) while simultaneously being viewed as a challenge to rebel authority over local populations (Wood & Sullivan, 2015). Further, rebels contest government territory or otherwise co-opt international aid to receive material benefits and maintain the loyalty of civilians where possible (Wood & Molfino, 2016). For rebel-controlled territories, rebels may also target aid workers directly, leaving these territories effectively inaccessible to aid workers (Harvey, 2013). Aid can also lead to conflict being exacerbated (Nunn & Qian, 2014), whether it be through refugee relief supporting militants (Lischer, 2003) or increasing uncertainty about the relative strength of either side (Narang, 2015). While in these studies aid is often found to spur conflict or violence, this is not consistent across studies.

Lyall (2019), for example, suggests that humanitarian aid can decrease violence because it can reduce local grievances and casualties, thereby reducing rebels' recruitment abilities, or incentivize locals to cooperate with the counterinsurgency. de Ree and Nillesen (2009) additionally find that aid reduces the risk of conflict onset in sub-Saharan Africa. Given that rebel authority is bolstered by aid provision in their territories (Carnegie et al., 2022a), whether

rebels view aid as a threat or a boon is also potentially dependent upon their governance tactics and incentives. Furthermore, sharp decreases in aid can increase the probability of conflict onset (Nielsen et al., 2011). These findings together create uncertainty in this proposed detrimental relationship but emphasize the incentives that both the government and rebels face when considering (preventing the) exploitation of foreign aid to influence the larger conflict. Given these considerations, this study provides insights by investigating potential mediating factors that might impact efforts on either side to leverage foreign aid to achieve strategic objectives. Further, this study investigates the impact of civil conflict and state influence on third party donors' potential ability to reach and maintain projects in localities that are in dire need of humanitarian support.

Natural Disasters and Foreign Aid

Governments also make strategic considerations regarding the acceptance and provision of disaster aid. Governmental influence can shape foreign aid responses, influencing what groups or regions get aid within the state's territory (Desportes, 2019), or even deny responses entirely as seen in the case of Myanmar's response to Cyclone Nargis in 2008 (Dany, 2020).

Authoritarian governments accept aid based in part on survival and need: how aid might spur or otherwise alter international and domestic pressure against their regimes (Paik, 2011).

Authoritarian states, therefore, may accept aid to use it as a heavily controlled tool to enhance state power or improve perceptions of the state, ensuring regime stability (Desportes and Hilhorst, 2020). This results in varying demands for foreign aid despite high humanitarian needs.

The supply of aid, however, is driven by both humanitarian needs and strategic factors. While aid increases post-disaster (Mogge et al., 2023; Becerra et al., 2014; Francken et al., 2012; Strömberg, 2007), the post-disaster aid literature is divided on whether humanitarian need is a

stronger influence on aid allocation than political or strategic interests (Mogge et al., 2023; Becerra et al., 2014) or if the latter may outweigh the former (Francken et al., 2012; Cheng and Minhas, 2021). Other factors can influence aid provision as well - media attention, for instance, can help provide larger inflows of aid (Becerra et al., 2012; Strömberg, 2007), contingent upon the type of disaster (Strömberg, 2007).

Because the literature highlights a potential tension between the strategic or political interests of the state and the humanitarian needs of the public, this study investigates how aid is dispersed and distributed within a given state to further illuminate the motivating factors that drive aid delivery. By looking at within-state variation in aid provisions, this study directly examines whether aid that country-level examinations would consider “post-disaster” is going towards localities stricken by natural disasters or is rather being dispensed elsewhere.

Natural Disasters and Conflict Dynamics

There is also a growing body of literature that explores the potential links between natural disasters and civil conflict dynamics. While this literature has grown, however, much of the current literature on how disasters may influence conflict present a debate about whether disasters are pacifiers of conflict onset (Slettebak, 2012; Caso et al., 2023; Ide, 2023a; Bergholt and Lujala, 2012) or instigators of conflict (Nelson, 2010; Ide, 2023a; Ide, 2023b; Bell and Keys, 2018; Nel and Righarts, 2008; Ide et al., 2021). Work investigating how disasters impact ongoing conflicts is more limited but demonstrates the importance of disasters in influencing both sides’ incentives and capabilities to conduct conflict.

For instance, natural disasters are found to increase the duration of conflict by decreasing state financial and military resources and damaging infrastructure, reducing state capacity for counterinsurgency, increasing the length of the war (Eastin, 2016). Much of the research on

disasters and ongoing conflicts also suggests a potential pacifying effect of disasters on conflict. For example, disasters can also inhibit insurgent capabilities to mobilize resources, preventing them from seizing opportunities that disasters may provide them (Eastin, 2016). Alternatively, de-escalation can also occur if both parties are significantly impacted by disasters (Ide, 2023). Similarly, Nemeth and Lai (2022) find that disasters improve the likelihood of conflict negotiations when both sides are negatively impacted by natural disasters, increasing the costs of short-term conflict.

However, other works indicate that disasters may spur violence in ongoing conflicts. For example, extreme variation in abundance or scarcity of rainfall can spur rebel or communal conflicts in various contexts (Eastin, 2018; Raleigh & Kniveton, 2012; Salehyan & Hendrix, 2014). Further, the negative impacts of disasters on states may require a shift in resources from counterinsurgency elsewhere, creating vulnerabilities that terrorists can exploit to conduct more attacks (Berebbi & Ostwald, 2011). Rebel strength may condition the influence of disasters on conflict, as strong rebels can take advantage of governments weakened by disasters, while weak rebel groups may focus on their own disaster response or are incapable of escalating their activities (Ide, 2023b).¹ Taken together with work suggesting a pacifying effect, the impact of natural disasters on conflict is shown to be dependent on how the natural disaster influences the balance of power. This paper's exploration of the spatial distribution of disasters across territorial control by different sides during a civil war helps provide a direct investigation of how disasters located in certain regions create varying effects in part because of the balance of power.

¹ Similar findings are also shown in considering water scarcity in civil conflicts (Keels, 2019).

The focus has been on the direct impact of disasters on government and rebel strength. However, the provision of aid after disasters can also influence combatant strength. As states face civil conflicts, they face a myriad of choices and priorities for how they should approach conflict whilst governing and attempting to provide some level of public goods to the citizenry at large as well as those who support the current leadership. While large-scale disasters can provide an opportunity to de-escalate conflict in some contexts (Nemeth & Lai, 2022; Ide, 2023b), the uptick in foreign aid in the aftermath of disasters (Mogge et al., 2023; Becerra et al., 2014; Francken et al., 2012; Strömberg, 2007) provides the state with an opportunity to further shift relative power in its favor – through winning the “hearts and minds” of the public (Sexton, 2016), rewarding supporters (Desportes & Hilhorst, 2020), or challenging rebel control (Wood & Sullivan, 2015). Even if the state may seek to take steps towards cooperation or de-escalation, it can still simultaneously aim to strengthen its own position in the conflict through foreign aid.

Theory

States facing conflict are faced with a unique set of incentives surrounding how to leverage foreign aid, especially after a natural disaster. I argue that conflict contexts impact government decision-making surrounding foreign aid in two ways: first, by incentivizing the state to consider aid strategically, pushing aid towards non-rebel controlled localities to prevent co-option or destruction of aid by rebel forces; and second, by exacerbating instability that could pose a risk to the government, incentivizing the state to deliver aid as reward and punishment to maintain power. Given these considerations, governments will simultaneously take both disasters and territorial control into consideration when determining foreign aid project locations. Further, given these mechanisms, I expect governments to prioritize aid based on whether the localities are within the government’s control or not first, and where relief is needed second, to maximize

assets and improve stability. I elaborate on each of these mechanisms and my subsequent expectations below.

Aid as a Strategic Item

Disasters during a civil conflict create a plethora of strategic and humanitarian challenges for states. Attempting to minimize suffering resulting from the disaster creates an inherent tradeoff with efforts spent on counterinsurgency (Eastin, 2016; Berrebi & Ostwald, 2011). Likewise, rebels also face a dilemma – prioritizing their own disaster response or, if they are capable enough, seizing an opportunity of state weakness (Ide, 2023b). However, foreign aid, especially developmental assistance, poses a potential strategic opportunity for both the state and the rebels, an opportunity that the state is disproportionately in control of shaping and wielding to its advantage. This control permits the state to act intentionally regarding allocation in conflict zones, especially when those zones have also been impacted by a natural disaster.

While rebels may exhibit influence in their territory, the state holds unequal sway over whether aid is dispensed in those territories to begin with. For example, in Pakistan’s displacement crisis in 2009, the government heavily constrained the humanitarian aid response such that the government effectively controlled the response completely (Harvey, 2013). For instance, the government limited access for UN humanitarian workers to affected populations to the degree that it “...completely shaped and constrained the humanitarian response in much of the affected area” (Cosgrave et al., 2010, 3) to the point that “...UN agencies did not work in an impartial, independent or neutral manner, but instead strongly supported one party to the conflict, the Government” (Cosgrave et al., 2010, 3-4). Many international actors seek to provide development assistance to help mitigate the costs of war for civilians but face heavy government involvement that can prevent such objectives from being (fully) realized.

While the interests of donors are relevant, the state holds heavy influence over where aid can be placed and has various means to pressure local project managers and aid providers into doing what the state wants. For example, an international non-governmental organization (INGO) representative working in Myanmar had their ID card with their photograph printed in a state newspaper article that stated their organization supported the “illegal and dangerous Rohingya” in Myanmar for delivering foreign aid in ways not condoned by the government (Desportes and Hilhorst, 2020). These types of pressures can lead to other INGOs or international organizations placing pressure on civil society organizations not to criticize the state too heavily if they fear punishment (Desportes and Hilhorst, 2020). Because of this, the state has both direct and indirect influence over disaster response through bureaucratic mechanisms such as the threat of expulsion, or even verbal and physical intimidation tactics for carrying out mandates in non-condoned ways, forcing responders to balance good relations with the state and fulfilling humanitarian needs (Desportes, 2019; Cosgrave et al., 2010). This influence over aid delivery gives the state the ability to spend aid money or distribute aid projects strategically rather than entirely based on humanitarian needs. Because of the power the state wields and the fungibility of aid resources, states will shift aid to regions they believe best help their cause politically and militarily.

Aid as a Tool for Stability

Disasters present an exogenous shock that might trigger popular contestation and the potential risk of regime change (Passeri, 2022). Natural disasters create considerable economic damage at a subnational level, depriving locals of income. Even if the state at large is unimpacted by a natural disaster, locally impacted regions may feel strong losses economically, making aid a relevant and valuable tool for offsetting losses (Kousky, 2014). This security risk is

further exacerbated in times of civil conflict. Natural disasters create additional instability, especially in times of weakness – weak institutions, regime control, or poor economic conditions are all factors that increase the vulnerability of states to the negative impact of natural disasters (Omelicheva, 2011). In conflict contexts, natural disasters can thus spark further instability, especially within individual localities. As both a preventative measure and a response to natural disasters, governments prioritize foreign aid and development to their own controlled territory.

Allowing the risk of co-optation or destruction means permitting the risk of further destabilization. Aid taken by the rebels means that much of the benefits of foreign aid that could have gone to the state are actively used against them. Being able to claim credit for aid-provided service delivery and development could change public perceptions surrounding government efficacy, for instance (Carnegie et al., 2022a). Disrupted aid cannot be utilized by the rebels, but it does not send the signal of competence and responsibility that states seeking further aid or either domestic or international support in the conflict desire. Further, the disruption of aid helps foment long-term grievances that could further lead to instability, harming counterinsurgency efforts. Development assistance in government-controlled territory also permits the state greater control over how it is wielded for and against those in the region.

Political interests can be powerful motivators for governments determining where to allocate post-disaster aid (Desportes and Hilhorst, 2020; Rosvold, 2020). Disaster responses can be used to achieve political ends, whether it be “...to end the protests in Ethiopia, to marginalize ethnic and religious minorities in Myanmar, and to assert the dominant party’s power in Zimbabwe” (Desportes and Hilhorst, 2020: 347-348). For instance, the 2014 Tokwe-Mukosi flood disaster in Zimbabwe saw resettlement result in improper care for victims of the flooding, with little in the way of permanent housing, compensation, or care (Hove, 2016). Flooding

victims were resettled either through economic coercion from the government, physical violence by the government to force victims to relocate, or self-settlement by the victims to areas that preserved their autonomy from government (Hove, 2016; Mucherera & Spiegel, 2021). Aid distribution is another such tool that can be utilized to help accomplish state interests without expending military power like more violent coercion tactics such as forced relocation (Desportes and Hilhorst, 2020). In the case of Zimbabwe, one form of economic coercion was threatening the end of relief aid to the afflicted if they did not relocate (Hove, 2016; Mucherera & Spiegel, 2021). Aid can thus provide both strategic incentives to control its placement while serving as a desirable stabilizing tool when in the government's control.

Even if the government is not directly in control of aid, such as when donors provide aid to those not directly under the recipient state's control (often referred to as "bypass aid") through groups such as NGOs (local or international), private actors, or multilateral organizations, recipient states can still receive political benefits from that aid. For instance, bypass aid can still change locals' preferences to make them less likely to challenge authoritarian governments while simultaneously creating conditions where governments can concede less to locals because of the aid they are receiving to address their grievances (DiLorenzo, 2018). The recipient government can also instill pressure and punishments on these alternative aid dispensers to shape where aid is delivered (Desportes, 2019). While governments may restrict access to areas for security considerations relevant to the safety of aid workers, security can also provide a plausible scapegoat to restrict the location of responses as it did in Pakistan (Cosgrave et al., 2010, 28). While many areas are impacted by conflict disproportionately, the state has a direct incentive to shape aid placement to regions and localities that are within its control. Through doing so, the state can attempt to bolster state capacity through aid development and resources or potentially

shift resources away from public good provisions towards the conflict now that they have supplementary support.

Considerations of Control

In countries facing a tradeoff between allocating aid towards conflict-impacted or disaster-impacted territories, the government has apparent reasons to consider both its strategic weight as well as its potential to help promote stability. I argue these considerations lead the government to prioritize its objectives related to defeating the rebels over other political and humanitarian concerns. More specifically, territorial control shapes the influence of a natural disaster's impact on how much foreign aid that territory may receive. Providing aid to government-controlled localities helps minimize the risk that aid is not destroyed or co-opted by rebel forces for their own usage or dispersion or to otherwise prevent the government from receiving the benefits. I expand upon these considerations by examining the two other options the state has for where aid should be allocated – areas that are in the rebels' control and areas that are contested territory.

There are a number of incentives for states to shift aid to non-rebel-controlled territories. First, areas that are under rebel control pose a threat to ensuring aid achieves the objectives the state aims for. Aid in rebel or contested territory could prove to mitigate domestic gains among the public (Carnegie et al., 2022a), counteract government narratives of competency (Desportes and Hilhorst, 2020), or otherwise enrich the insurgents (Wood & Sullivan, 2015; Addison et al., 2002). Providing aid in areas that are firmly within their control, however, ensures that the state can attribute the aid to their own actions and good management of resources while attempting to signal good governance and care for the public.

Second, natural disasters that impact rebel-controlled territory provide an opportunity for the government to strike against them. For instance, natural disasters create funding losses for rebels reliant on natural resource production because of rapid-onset disasters destroying such sources of income (Tominaga & Lee, 2021) as well as potentially weakening rebel recruitment efforts through weakening organizational structure and supply lines (Walch, 2018). In the case of Nigeria, flooding had aided government forces in accessing Boko Haram territory (Ide, 2023b); for the Philippines, natural disasters have also been shown to aid in efforts to change territorial control (Zhou, 2025). These reasons provide incentives against sending aid, but aid in rebel territory provides advantages to the government as well which may offset the risks described above.

While the state has incentives to send aid elsewhere, there are reasons to provide aid even in rebel territory. Facing a natural disaster often increases the amount of media attention the country and the impacted area receive from those abroad, depending on the type of disaster (Strömberg, 2007). This creates an opportunity for the state to signal that they are actively working for civilian interests and care about their humanitarian needs to maximize the benefits of international attention, whether it be to receive aid or otherwise bolster legitimacy. Additionally, supporting foreign aid projects in rebel territory provides an opportunity for the government to reestablish its presence in the locality through aid provisions as well as to track rebel recruitment patterns and strategy (Walch, 2018). Given these considerations, I argue that aid allocations to impacted rebel-controlled territory may be reduced as a means of balancing the benefits of benefits of aid with the potential risk in rebel territory.

While some territory can be considered strongly held, uncontested government or rebel-controlled areas, control within a single territory may be heavily divided between combatant

parties. These contested territories pose a dilemma for all parties involved. First, foreign aid providers and workers may seek to avoid (or are prevented by the state from going to) areas facing active contestation or conflict out of safety concerns for their staff and projects. As mentioned previously, governments may at times restrict access to areas because of security considerations, providing another barrier for those who want to provide aid (Cosgrave et al., 2010). These areas see more active conflict than uncontested territory, and parties on both sides have active stakes in ensuring that they can either receive any tangible benefits of aid or otherwise prevent their enemy from receiving those benefits.

Second, while the government has a chance to extract benefits from foreign aid in rebel territory, rebels also can extract benefits such as enhancing their legitimacy (Carnegie et al., 2022a) from the process as well. Contested areas do not provide this clear benefit to rebels and may be regions where rebels are more likely to consider aid attempts as challenges to their authority in that region and otherwise as opportunities to extract resources that can help shift the power dynamic back in their favor (Wood & Sullivan, 2015; Addison et al., 2002). Undermining rebels in contested territory may thus serve as a strategy that sparks further conflict, disincentivizing the state from providing aid to these regions (Sexton, 2016). Because contested territory results in unclear or more risk-laden benefits for both sides, it becomes less desirable to be chosen as an aid target as a result.

Third, aid in areas facing active contestation or conflict may still be more likely to be destroyed, looted, or otherwise rendered or determined to be unsafe for aid workers to operate within, even if the rebel group in question is much weaker than the government. For instance, in Afghanistan, government-provided counterinsurgency aid reduced insurgent violence in government-controlled areas while contested areas saw increases in violence (Sexton, 2016). The

reasoning is clear: insurgents have limited resources and therefore choose targets for attacks carefully to minimize costs and retaliation while preventing government infrastructure from building up that can “win the hearts and minds” of the public (Sexton, 2016). Because of these considerations, I anticipate that actively contested territory is likely to receive the least amount of aid, whether it be to preserve the safety of aid projects and their workers or to maximize the value to the state that these projects may provide. Given the reasons described above, I summarize my empirical expectations in the hypotheses and table below:

Prioritization Order for Foreign Aid		State of Territorial Control		
Impacted by Natural Disaster		Government Control	Contested Territory	Rebel Control
	Impact	High	Lowest	Medium
	No Impact	Medium	Lowest	Low

Table 1: Summarization of Hypotheses 1, 2, & 3 in tandem with colors indicating government priority from red (lowest priority) to blue (highest priority).

H1a: Rebel-controlled areas that are impacted by natural disasters will be less likely to receive foreign aid than government-controlled areas that are impacted.

H1b: Rebel-controlled areas that are not impacted by natural disasters will be less likely to receive foreign aid than government-controlled areas that are not impacted.

H2: Government-controlled areas that are impacted by natural disasters will be more likely to receive foreign aid than government-controlled areas that are not.

H3: Contested territories will be less likely to receive foreign aid than either rebel-controlled or government-controlled areas.

Methodology

The Philippines as a Case of Conflict, Disasters, and Foreign Aid

In 2018, 63 percent of official development assistance went to fragile or conflict-impacted states (Desai, 2020). I examine the case of the Philippines as a starting point to understand how conflict-impacted states disperse foreign aid in the wake of a disaster.

The Philippines has faced civil conflict since the late 1960s from a number of groups that have entered and exited the conflict over the years. In particular, during the time frame of this analysis, the Philippines faces challenges from multiple groups, such as the Communist Party of the Philippines/the New People's Army (CPP/NPA), the Moro Islamic Liberation Front (the MILF), the Moro National Liberation Front (MNLF), and the Abu Sayyaf Group (ASG), among others. The CPP/NPA have been the Philippines' most persistent opponents, having been engaged in conflict against the state since 1968. In just the conflict with the CPP/NPA alone, over 40,000 lives have been lost (Engelbrecht, 2024); conflict in Mindanao has resulted in the deaths of over 120,000 and the displacement of millions (International Crisis Group, 2024). As the state has balanced priorities along multiple conflict fronts at various points in time, the state has also faced immense humanitarian pressures from its environment as well.

The Philippines faces frequent natural hazards, creating persistent tests of its infrastructure and resiliency at various scales. The Philippines is extremely at-risk from natural disasters, having lost over 70,000 lives since 1990 to over 560 disasters (World Bank, n.d.). Many of these disasters have seen a pronounced international response, such as Typhoon Haiyan (Field, 2018). The Philippines also faces a high amount of small-scale natural hazards as well, creating further impacts to localities, facing 225 small-scale natural hazard incidents in just 2024 alone (Del Prado, 2025). In the wake of Supertyphoon Yolanda, the CPP released a statement

appealing to foreign governments and international aid agencies to dispense aid directly to local organizations rather than via the government (NDFPIO, 2014). In their statement, they argued such action would help ensure that civilians receive foreign aid, accusing the government of ‘plundering’ aid and selling aid rations in Manila while rural areas receive nothing and intensifying combat operations in disaster-impacted areas (NDFPIO, 2014).

I focus the scope of my analysis on the Philippines from 2011 to 2014. There are two main reasons why this provides a beneficial scope that can provide a useful contribution to future research. First, a growing body of research has examined the influence of foreign aid in the Philippines (Rosvold, 2019; Han, 2025; Field, 2018; Montinola et al., 2020; Trinidad, 2019). This study aims to contribute to this literature by examining how territorial control may serve as a conditional factor for the placement of foreign aid in the aftermath of a natural disaster in the case of the Philippines, helping detangle the relationship between aid and conflict (Findley, 2018). By doing so, this study builds on qualitative evidence that suggests that this dynamic may be in place (Field, 2018) and helps potentially validate quantitative evidence that the Philippines may impact aid flows from prominent organizations like the World Bank (Rosvold, 2019). Further, while this study focuses on aid from the World Bank, future studies may seek to investigate how state-provided aid may be more or less resilient to such dynamics as described here, given that state relationships have been shown to impact the prevalence of aid (Trinidad, 2019).

Second, examining a single-country case rather than broader cross-national trends over a large period of time aids in testing underlying causal mechanisms and theories that would be more difficult to parse from a larger scope of cross-national data (Gerring, 2004). Given this, the Philippines serves as a useful case to examine how foreign aid is shaped by what governments

view as a priority. Rosvold (2019), for instance, highlights that World Bank aid disbursement in the Philippines is likely subject to political favoritism, as areas where politically dominant ethnic groups are located are more likely to receive aid, even if other areas may need it more.

Furthermore, Field (2018) highlights that the Philippines has previously faced challenges in aid allocation between disasters and conflict, with evidence suggesting that disaster-impacted territory took precedence over conflict-impacted territory for the allocation of international aid. Because of this, better understanding the dynamics at play in the Philippines during conflict may help inform future work in other civil conflict contexts.

Data and Variable Description

To test my hypotheses, I use data at the municipality-month level of analysis for the Philippines from 2011 to 2014, excluding the island of Mindanao, originally sourced from Zhao (2025), with a sample size of 29,200 municipality-months. The municipality serves as the second smallest administrative unit in the Philippines, included in geocoded data projects for foreign aid and natural disasters, making it a viable means to see how foreign aid placement is influenced by local conflict and environmental dynamics. Using replication materials from Zhao (2025), this study focuses upon the conflict between the CPP/NPA and the Philippines government. During this time period, the CPP/NPA is the most active combatant. According to the UCDP Georeferenced Event Dataset v. 25.1 (GED) (Davies et al., 2025; Sundberg et al., 2013), the CPP are involved in over a majority of conflict events from 2011-2014, making up 67% of government-rebel battles and the perpetrator of 43% of cases of violence against civilians during the same period.

For my dependent variable, I use the World Bank Geocoded Research Release, Version 1.4.2 from AidData (2017) for this analysis, which includes all World Bank aid projects from 1995 to 2014. In the case of the World Bank, evidence showcases that donors such as the United States play a meaningful role in shaping aid commitment and disbursement through both formal and informal channels (Fleck & Kilby, 2006; Kilby, 2013). However, aid recipients still have meaningful influence. For instance, states with temporary UN security council membership receive a higher amount of World Bank projects, although loan amounts remain consistent (Dreher et al., 2009). Further, existing evidence highlights that domestic alliances in recipient states, such as the Philippines, can still shape aid delivery subnationally despite donor policies over how aid can be used (Rosvold, 2020). For the Philippines, these aid projects go through the government for their dispersion (Rosvold, 2020), making it suitable for testing. As a result, examining whether foreign aid delivery is also influenced in the Philippines by the state of territorial control provides meaningful insight into the role of recipient influence over multilateral aid delivery through organizations such as the World Bank.

To help ensure that the projects listed are geographically within the municipality they would be associated with, I follow previous research and do not include aid projects that are geocoded to more than 25 kilometers of the correct location to have higher confidence that projects are being properly attributed to the correct municipality (Briggs, 2021; Briggs, 2018). I create several measures from this process. First, I create an indicator variable for whether or not there is a World Bank-funded local aid project in each municipality in a given month. Second, I create an average of *aid disbursements per month* for each project and aggregate them to the municipality-month level. For robustness, I also create an estimate for aid commitments per

month as well as the number of aid projects in each municipality in a given month for further analysis.²

To measure the presence of a natural disaster in a given territory, I first rely on replication data sourced from Zhao (2025), which codes the *presence of a natural disaster* in a municipality if the level of precipitation deviates substantially from normal. These data are originally sourced from the Integrated Multi-satellite Retrievals for GPM dataset (IMERG), version 6.0. This measure captures the most common incidences of natural disasters, such as landslides, floods, and storms, and is consistently available and can be considered accurate for each municipality across the time period.³ While this data covers the most frequent disasters, it does not cover the whole scope of potential disasters that can occur. For instance, the Bohol earthquake in 2013 resulted in over two hundred deaths and significant destruction, resulting in the Philippines calling for international aid, contributing to the state's aid allocation dilemma (Field, 2018). Because of this, I conduct further analysis using an alternative data source for robustness.

To measure the level of territorial control in a given municipality, I rely on Zhao's (2025) data on yearly estimates of the *percentage of each municipality controlled by the CPP/NPA*. This data is sourced from Armed Forces of the Philippines (AFP) intelligence assessments, meant for internal use, and coded the degree to which the NPA was present in each barangay (Zhao, 2025). This measure at the barangay level is then aggregated to a municipality level, in which higher

² To take total disbursements to the municipality-month level, I follow previous research (e.g., Briggs, 2021) and average the total disbursement to the project over the course of its duration. For example, if a municipality has one project with 1000 dollars in total disbursements, and it lasts for 10 months, then each municipality-month has an average commitment per month of 100 dollars for those 10 months.

³ Specifically, precipitation that falls at or above 1.6 on a standardized scale is coded as the presence of a natural disaster in that given municipality-month. For more details on the scale, see Zhao (2025, p. 14-15).

percentages indicate more barangays within the municipality are controlled by the CPP/NPA. Most municipalities exist without the CPP/NPA's presence throughout the time period, with only about 13.36% of the municipality-months facing some level of contestation.

Finally, I control multiple factors that could confound the relationship between my key variables. Because of the limited data available at the municipality level, I incorporate a number of controls from the PRIO-GRID version 2.0 static and yearly datasets (Tollefson et al., 2012). Using GIS software, I match grid-cells to municipalities and subsequently aggregate grid-cell level data to the municipality level. In particular, I include estimates of the average *distance to the capital* (Weidmann, 2010), the highest estimated number of *excluded ethnic groups* in the municipality (Vogt et al., 2015), the average maximum *nighttime light intensity* (from the DMSP-OLS Nighttime Lights Time Series Version 4)⁴, the average level of *forest ground cover* (Bontemps et al., 2009), level of *shrub ground cover* (Bontemps et al., 2009), whether or not there was a *natural resource deposit* in that location (Lujala, 2007; Gilmore et al., 2005; Lujala et al., 2005; Balestri, 2015; Balestri, 2012; Lujala, 2009), and the average maximum *infant mortality rate* (Storeygard et al., 2008; CIESIN, 2005) at a yearly level for each municipality. Controls such as these help account for potential differences in pre-existing levels of infrastructure or humanitarian need outside of the effects of natural disasters and territorial control.

I also include a measure of the level of conflict-related violence in each municipality over time. I use UCDP's Georeferenced Event Dataset (GED) Global version 24.1 (Sundberg &

⁴ Because the maximum ethnic group and nighttime light intensity variables are missing for the year 2014, I code these values for 2014 with each municipality's value for the previous year. For robustness, I also conduct my analysis with the original coding, leaving those observations as missing. I do not find a meaningful difference in the findings as a result of this choice.

Melander, 2013), which includes information on an estimate of how many individuals died on any side of the event (e.g., how many soldiers, rebels, or civilians died in that given event). Using GIS software and these event-level data, I aggregate all battle fatalities within a municipality-month and create a count variable, the *Number of Conflict Deaths*, which captures the total number of rebel, government, and civilian deaths that occurred within that municipality-month. To account for the potential long-term temporal dynamic between local foreign aid provisions, the nature of the conflict, and natural disasters, I include year-level fixed effects. Summary statistics for all key variables can be found in Table A1.

Modeling and Results

To test my hypotheses, I run a set of logistic regressions where the dependent variable is the *presence of an aid project*. Because the length of time between a natural disaster and foreign aid arriving in a given locality may vary, I estimate four models, each with a different window for when a natural disaster took place: first, where the presence of a natural disaster is coded as 1 if there was a disaster in that month, and 0 otherwise; second, where the presence is coded as a 1 if there was a disaster in the last four months, including the present month; the third and fourth models use a disaster window of 8 months and 12 months, respectively. Because of the interaction between natural disasters and territorial control, I investigate the relationship between these components and foreign aid by constructing predicted probability plots to model the relationship between natural disasters over varying levels of control.

Table 2 - Influence of Disasters & Control on Presence of Aid Projects

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Disaster (1 Month)	0.052			

	(0.039)			
Disaster (4 Months)		0.012 (0.028)		
Disaster (8 Months)			-0.040 (0.030)	
Disaster (12 Months)				-0.036 (0.035)
Rebel Control (%)	0.057*** (0.004)	0.054*** (0.005)	0.050*** (0.006)	0.040*** (0.007)
Rebel Control ²	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Disaster (1 Month) × Rebel Control	0.010 (0.013)			
Disaster (4 Months) × Rebel Control		0.012 (0.008)		
Disaster (8 Months) × Rebel Control			0.016 ⁺ (0.008)	
Disaster (12 Months) × Rebel Control				0.028** (0.009)
Disaster (1 Month) × Rebel Control ²	-0.000 (0.000)			
Disaster (4 Months) × Rebel Control ²		-0.000* (0.000)		
Disaster (8 Months) × Rebel Control ²			-0.001** (0.000)	
Disaster (12 Months) × Rebel Control ²				-0.001*** (0.000)
Avg. Distance to Capital	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Max. # of Excluded Ethnic Groups	-0.197*** (0.033)	-0.198*** (0.033)	-0.202*** (0.033)	-0.203*** (0.033)
Avg. Maximum Nighttime Lights	0.676* (0.338)	0.689* (0.339)	0.758* (0.340)	0.733* (0.339)
Avg. Forest Ground Cover	0.014*** (0.002)	0.014*** (0.002)	0.014*** (0.002)	0.014*** (0.002)
Avg. Maximum Infant Mortality Rate	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Avg. Shrub Ground Cover	-0.171*** (0.009)	-0.171*** (0.009)	-0.171*** (0.009)	-0.172*** (0.009)
Natural Resource Deposit	-0.224*** (0.029)	-0.224*** (0.029)	-0.221*** (0.029)	-0.223*** (0.029)

Conflict-Related Deaths	0.197** (0.071)	0.198** (0.071)	0.198** (0.071)	0.196** (0.071)
Constant	-0.574*** (0.088)	-0.573*** (0.088)	-0.550*** (0.089)	-0.548*** (0.090)
Observations	29200	29200	29200	29200
Pseudo R-Squared	0.03	0.03	0.03	0.03

Standard errors in parentheses

+ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

Looking first at Figure 1, I find mixed results in relation to my hypotheses. First, I find support for Hypotheses 1a and 1b: rebel-controlled areas are less likely to receive foreign aid than government-controlled areas. However, I find that there is no difference between impacted and unimpacted locations within a certain level of government control; regardless of the additional humanitarian need that may be present due to the disaster in one area compared to the other, the probability of receiving aid is the same between them. Similarly, contested areas are the most likely to receive aid, rather than areas entirely within government or rebel control, contrary to Hypothesis 3. However, when the window of relevancy expands to 8 or 12 months in Figure 2, while H1a and H1b remain supported, I find that unimpacted areas are more likely to receive World Bank aid than impacted areas within contested municipalities. These results suggest that the value of winning the “hearts and minds” within contested regions is ultimately a higher priority than the potential risk for destruction or co-optation within contested municipalities. However, as contested areas become increasingly rebel-controlled, the likelihood quickly drops to nearly 0, dropping by nearly 25% for every 10 percentage point increase in rebel control once rebels become the majority in control.

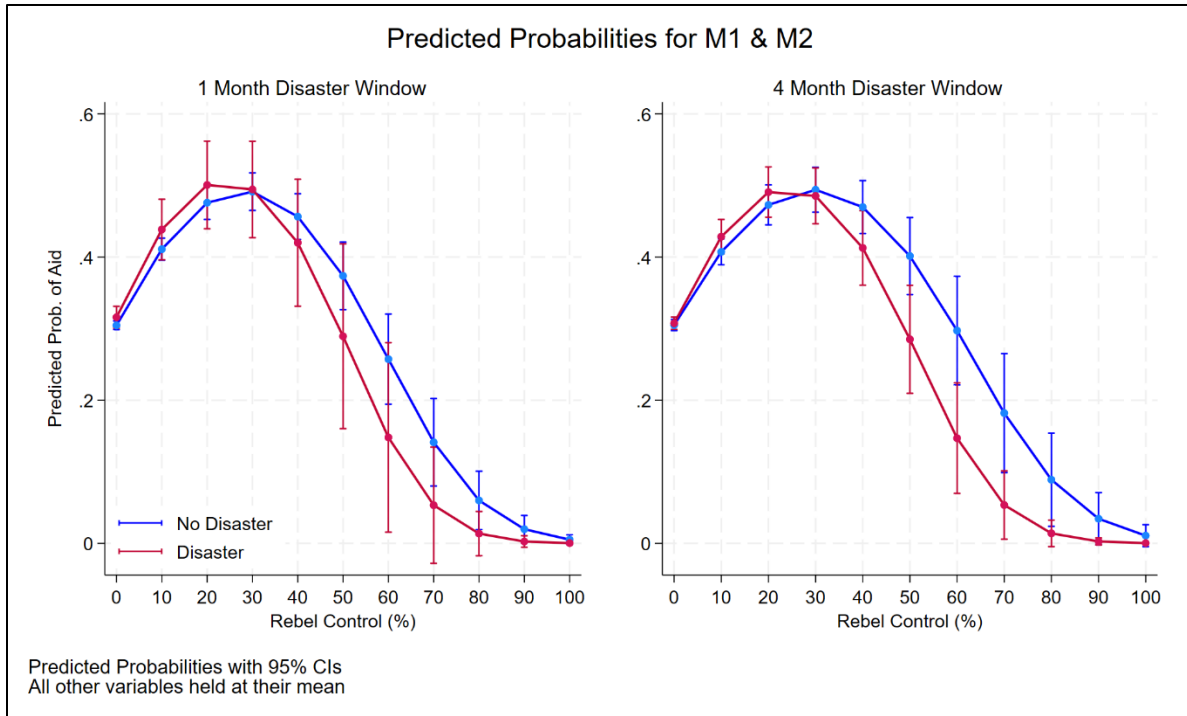


Figure 1: Predicted probability plots for the likelihood of local aid project presence, based on Models 1 and 2 of Table 2.

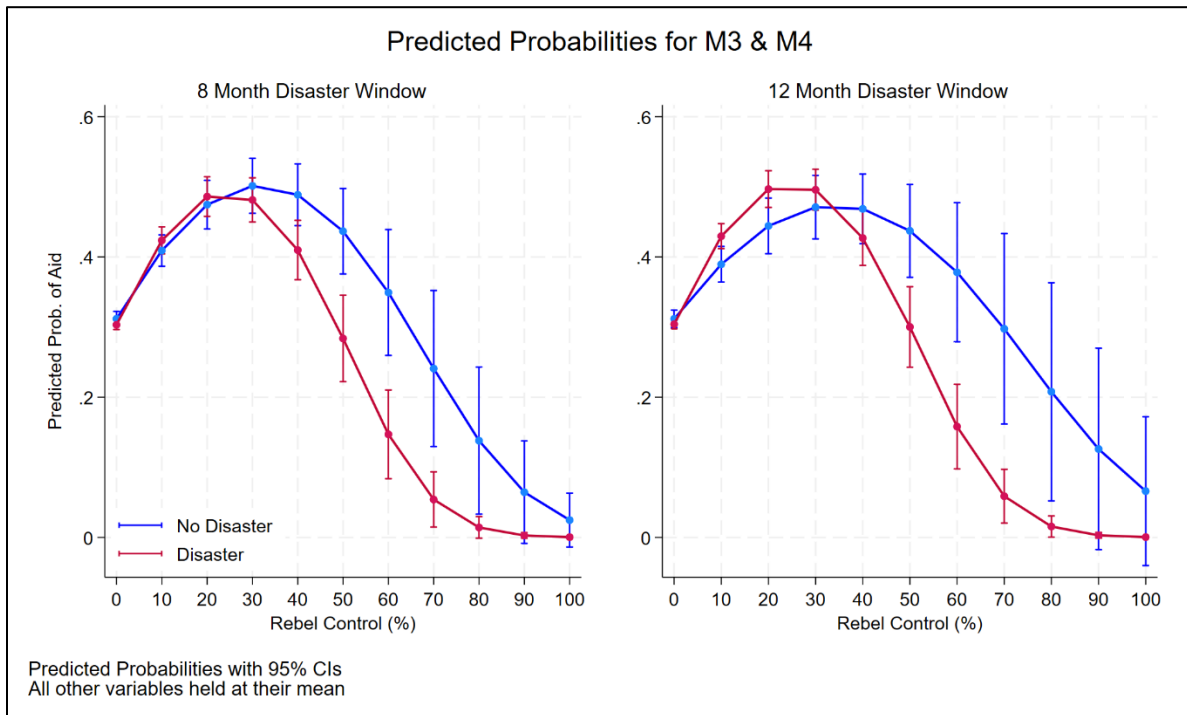


Figure 2: Predicted probability plots for the likelihood of local aid project presence, based on Models 3 and 4 of Table 2.

Looking briefly at the controls, I find no effect for the average distance of a municipality to the capital. However, as the maximum number of excluded ethnic groups increases, the likelihood of an aid project decreases, similar to findings by Rosvold (2019), who finds that aid is more likely to go where politically dominant ethnic groups are located. Interestingly, I find that municipalities with natural resource deposits are less likely to receive World Bank aid, even after accounting for where conflict events happen and where the CPP/NPA have some level of control, potentially indicating that these areas are in less need of aid or that the presence of the resource discourages aid placement in these localities. Lastly, I find that as the average level of nighttime light emissions increases and shrub cover declines, World Bank aid becomes more likely to be present within the municipality, indicating that aid is likely to go to municipalities with higher levels of infrastructure or development rather than regions that are less urbanized.

To further investigate the relationship between disasters, territorial control, and foreign aid, I run a series of robustness checks. First, I use an alternative source to measure natural disasters using data provided by the Geocoded Disasters (GDIS) dataset (Rosvold and Buhaug, 2021a; Rosvold and Buhaug, 2021b).⁵ When comparing to disaster classification using GDIS, I find that support for H1a and H1b remains; government-controlled areas receive consistently more support than rebel-controlled areas across multiple windows of relevancy for the natural disaster. While I find some potential for disaster-impacted areas in contested zones to be more likely to have a World Bank aid project, the results overall indicate that there is little difference in treatment between disaster-impacted and unimpacted areas in whether they receive foreign aid

⁵ This dataset takes natural disasters data from the Emergency Events Database (EM-DAT) (Guha-Sapir et al., 2014) and identifies the geographic location of the events. I manually match the names of locations specified in GDIS to municipalities from Zhao (2025), coding a municipality-month as a 1 if a natural disaster took place in that municipality, and a 0 if not. If there was a disaster that impacted a higher-level administrative unit (e.g., a region in the Philippines), that disaster was coded as a 1 for all constituent municipalities within that unit.

or not. Rather, zones of contestation between the government and CPP hold meaningful weight over the likelihood of receiving World Bank aid. I again find consistent support for the most contested areas to be the most likely to receive foreign aid, especially compared to entirely rebel-controlled regions.

Second, I run a set of panel OLS regressions with municipality-level random effects in which the dependent variable is the average aid disbursements per month to investigate whether the *amount* of funding also varies over control and disaster impact. I find that disbursements vary more extremely based on territorial control in disaster-impacted localities. Disaster-impacted localities see a more prominent U-shaped relationship between territorial control and predicted aid disbursements per month than non-impacted localities. Areas that are majority-controlled by the government are predicted to receive thousands more in aid disbursements per month than in majority-rebel controlled regions. This pattern is also consistent when estimated via a two-way fixed effects (TWFE) model up to the 8-month window. In conjunction with the previous findings, this may indicate that projects in contested areas are more frequent and larger in scale than projects in entirely government-controlled areas, while projects in predominantly rebel-controlled areas are both unlikely and less funded when they do occur.

Third, I also create a measure of natural disasters based only on where coordinate points provided by GDIS are located, where only municipalities overlapping with the provided coordinates are coded as having been impacted by a natural disasters. Fourth, I conduct the analysis using time-cubic polynomials rather than year-fixed effects. Fifth, I include region-level fixed effects to account for potential spatial effects that were otherwise uncaptured by the initial model specifications. Sixth, I conduct the analysis at the municipality-year level; in this analysis, the window of relevancy is whether there was a disaster in the previous year. Lastly, I conduct

the analysis while clustering the standard errors by municipality. Tables and figures for these checks can be found in the Supplementary Materials. The results across these checks are largely consistent across multiple different windows of relevancy for a given disaster: rebel-controlled regions are less likely to receive World Bank aid than government-controlled regions, and as contestation begins to favor the rebels, the likelihood of aid declines.

Discussion and Conclusion

Looking to the Philippines' case for insight provides information that may generalize to other cases more broadly. Like many civil conflicts, much of the time the Philippines has spent in conflict, including the timeframe examined here, has been at a minor intensity, with less than 1,000 battle-deaths in a given year (Gleditsch et al., 2002; Davies et al., 2025). This lower intensity may influence aid placement, in that states may be more willing to permit aid in contested territories when conflict is not high, as the risk of destruction may be lower than in times of total war. Further, the Philippines experiences conflict with multiple parties with a variety of ideological and/or separatist ideals. While this study only examines control from the CPP/NPA, future research may seek to examine whether the moderating influence of control on foreign aid is exacerbated in separatist dyadic contexts, given the strategic weight that the state may place on preventing separatist movements from growing (Walter, 2006).

However, there are some considerations as to the uniqueness of the case. First, the Philippines has existed in a state of civil conflict for multiple decades with groups like the CPP/NPA among others, and so the dynamic captured in the brief timeframe analyzed is one with years of history and information exchanged through negotiations and battle. Conflicts that have just begun or are short-lived may have a different moderating relationship between

territorial control and foreign aid. Further, the Philippines has engaged in a strategy of containment with groups like the CPP/NPA at various points during its conflict, while many other conflicts may be in states of total war, for instance (Staniland, 2021). The Philippines is also a high-frequency case when it comes to natural disasters, ranking high on the World Risk Index (González, 2024). Further analysis on low-frequency cases where disasters are uncommon, or where high-intensity disasters are less frequent, may prove to find different results.

This analysis indicates that while disaster impact may influence who receives aid, territorial control considerably influences the distribution of where foreign aid is placed subnationally in the context of the Philippines. In addition, I find that within territories firmly controlled by the government, there is no statistically significant difference in the likelihood of aid placement or amount of aid disbursements per month between disaster-impacted and non-impacted territories. This raises a puzzle for future scholars to further explore in other cases. It could be that the findings from Rosvold (2020) and Öhler and Nunnenkamp (2014) are the answer, in which the primary consideration within government-controlled territory is driven less by humanitarian need than by political needs for World Bank aid. Given the findings are tied to World Bank aid, future scholars may seek to conduct similar analysis, incorporating territorial control into their analysis, using UN disaster aid as a comparison given that current findings reflect humanitarian needs take precedent in such cases (Dellmuth et al., 2021).

Lastly, I consistently find that contested municipalities are the most likely and most funded places for World Bank aid, further indicating that states may seek to leverage foreign aid as a tactic to win the hearts and minds of the public or otherwise respond to counterinsurgency. Future analysis of the influence of disasters, foreign aid, and conflict, may benefit from considering how factors that may influence a government's willingness to risk the loss or co-

optation of aid, such as its pre-existing capacity, legitimacy, and presence in local communities, impacts the likelihood for aid in contested localities.

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Appendix

Table A1 – Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Presence of Project	77952	.33	.47	0	1
Avg. Monthly Aid Disbursements	77952	2630.614	9822.858	0	208041.11
# of Aid Projects	77952	.403	.636	0	4
Avg. Monthly Aid Commitments	77952	6585.395	34729.984	0	711291.14
IMERG Disaster (Present)	77952	.085	.278	0	1
IMERG Disaster (Within 4 Months)	77952	.277	.448	0	1
IMERG Disaster (Within 8 Months)	77952	.459	.498	0	1
IMERG Disaster (Within 12 Months)	77952	.584	.493	0	1
Rebel Control (%)	77952	2.672	9.523	0	95.238
Avg. Distance to Capital	77424	468.678	295.726	29.764	1085.302
Max. # of Excluded Ethnic Groups	58068	.839	.601	0	2
Max. # of Excluded Ethnic Groups (Extended Sample)	77424	.839	.601	0	2
Avg. Average Nighttime Lights	38712	.089	.045	.061	.241
Avg. Maximum Nighttime Lights (Extended Sample)	40325	.089	.045	.061	.241
Avg. Forest Ground Cover	77424	15.331	10.824	.042	57.744
Avg. Maximum Infant Mortality Rate	77424	340.651	92.547	197	507
Avg. Shrub Ground Cover	77424	1.707	1.812	.003	9.466
Natural Resource Deposit	77424	.414	.493	0	1
Deaths from Conflict (Unknown In	77952	.022	.938	0	236
GDIS Disaster (Present)	77952	.184	.387	0	1
GDIS Disaster (Within 4 Months)	77952	.506	.5	0	1
GDIS Disaster (Within 8 Months)	77952	.75	.433	0	1
GDIS Disaster (Within 12 Months)	77952	.871	.336	0	1
GDIS Disaster (Coords) (Present)	77952	.01	.098	0	1
GDIS Disaster (Coords) (Within 4 Months)	77952	.087	.282	0	1
GDIS Disaster (Coords) (Within 8 Months)	77952	.203	.402	0	1
GDIS Disaster (Coords) (Within 12 Months)	77952	.268	.443	0	1
Province on Mindanao	77952	.276	.447	0	1

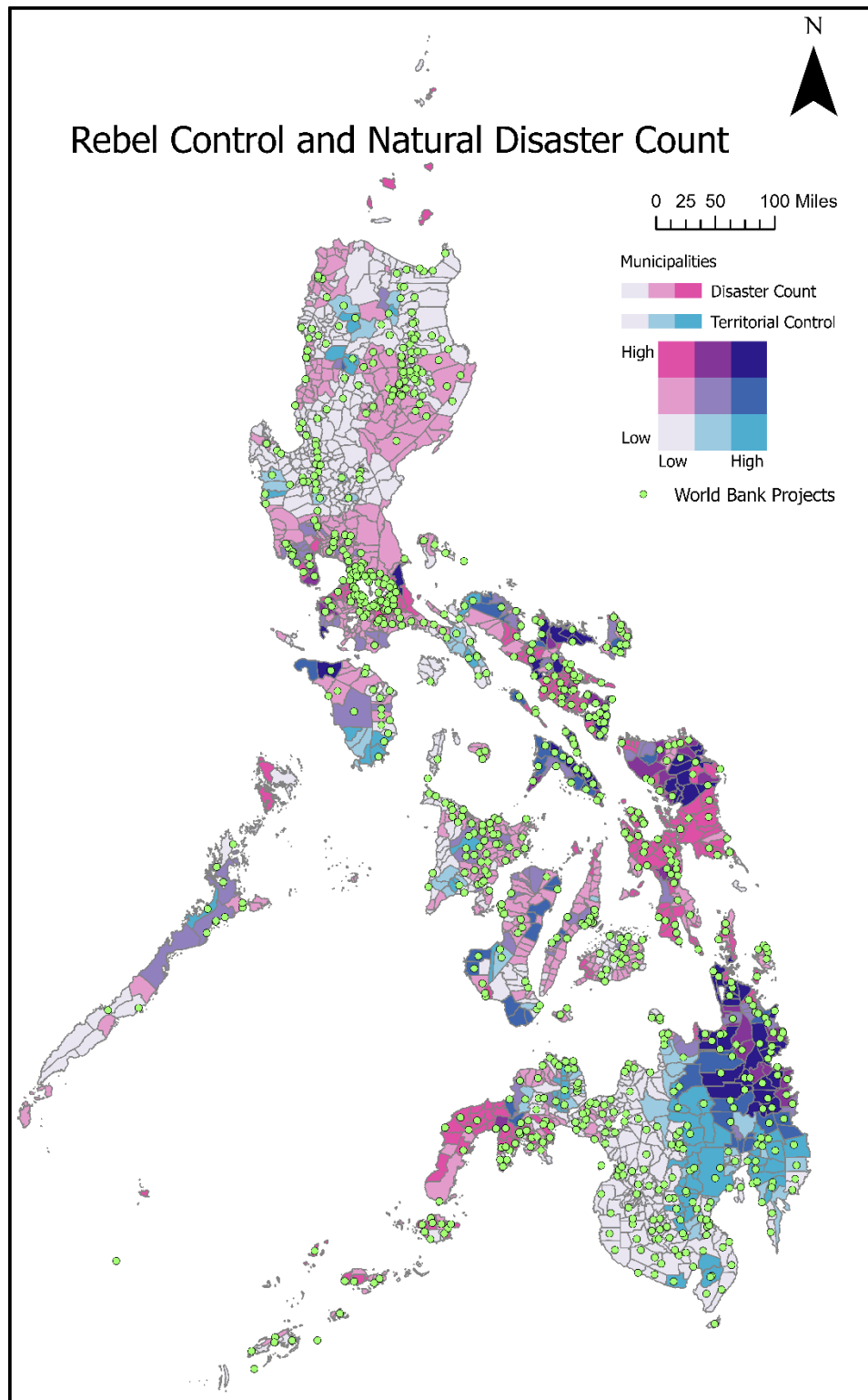


Figure A1: Choropleth and Point Map of World Bank Projects in Relation to CPP Territorial Control and Natural Disasters in the Philippines

Table A2 - Influence of Disasters (GDIS) & Control on Presence of Aid Projects

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Disaster (1 Month)	-0.010 (0.034)			
Disaster (4 Months)		-0.025 (0.029)		
Disaster (8 Months)			-0.094* (0.037)	
Disaster (12 Months)				0.059 (0.049)
Rebel Control (%)	0.059*** (0.005)	0.067*** (0.006)	0.065*** (0.009)	0.069*** (0.012)
Rebel Control ²	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Disaster (1 Month) × Rebel Control	-0.008 (0.010)			
Disaster (4 Months) × Rebel Control		-0.017* (0.008)		
Disaster (8 Months) × Rebel Control			-0.011 (0.010)	
Disaster (12 Months) × Rebel Control				-0.013 (0.013)
Disaster (1 Month) × Rebel Control ²	0.000 (0.000)			
Disaster (4 Months) × Rebel Control ²		0.001** (0.000)		
Disaster (8 Months) × Rebel Control ²			0.000+ (0.000)	
Disaster (12 Months) × Rebel Control ²				0.000 (0.000)
Avg. Distance to Capital	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Max. # of Excluded Ethnic Groups	-0.196*** (0.033)	-0.196*** (0.033)	-0.192*** (0.033)	-0.199*** (0.033)
Avg. Maximum Nighttime Lights	0.714* (0.337)	0.758* (0.338)	0.792* (0.339)	0.685* (0.338)
Avg. Forest Ground Cover	0.014*** (0.002)	0.014*** (0.002)	0.014*** (0.002)	0.014*** (0.002)
Avg. Maximum Infant Mortality	-0.000	-0.000	-0.000	-0.000

Rate	(0.000)	(0.000)	(0.000)	(0.000)
Avg. Shrub Ground Cover	-0.172**	-0.172**	-0.172**	-0.171**
	(0.009)	(0.009)	(0.009)	(0.009)
Natural Resource Deposit	-0.224**	-0.225**	-0.227**	-0.223**
	(0.029)	(0.029)	(0.029)	(0.029)
Conflict-Related Deaths	0.197**	0.198**	0.198**	0.196**
	(0.071)	(0.071)	(0.072)	(0.071)
Constant	-0.569**	-0.554**	-0.474**	-0.634**
	(0.089)	(0.091)	(0.096)	(0.101)
Observations	29200	29200	29200	29200
Pseudo R-Squared	0.03	0.03	0.03	0.03

Standard errors in parentheses
⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

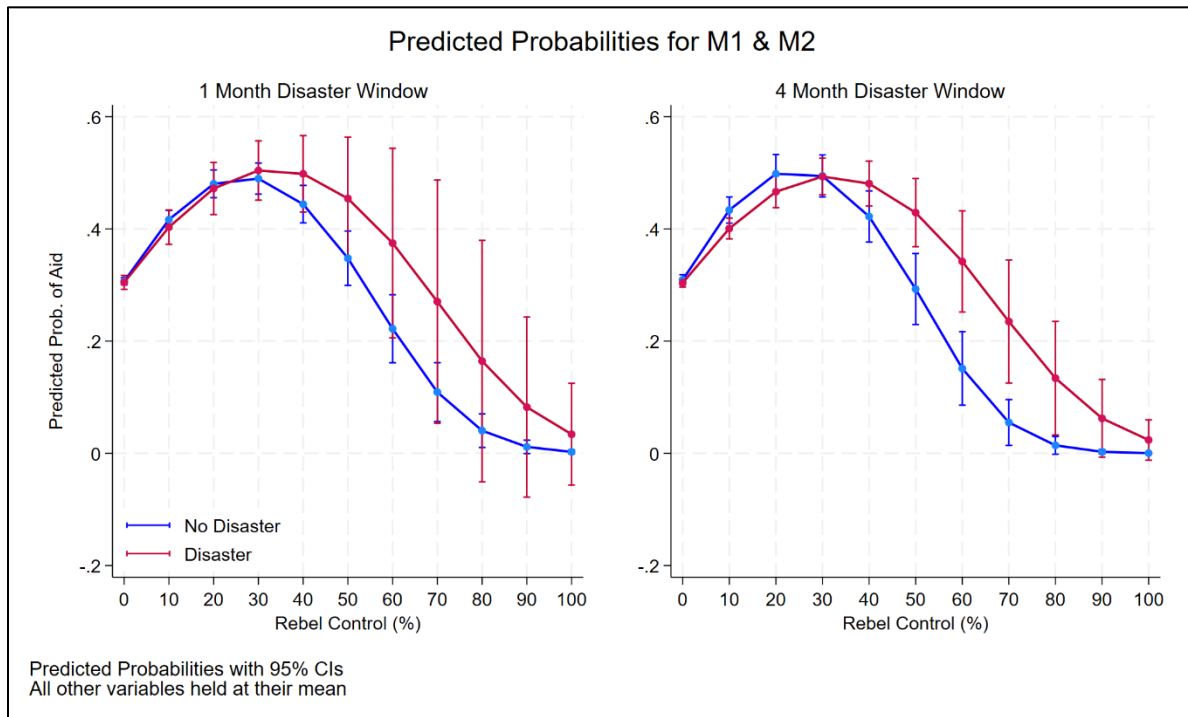


Figure A2: Predicted probability plots for the likelihood of a local aid project being installed, based on Models 1 and 2 of Table 3.

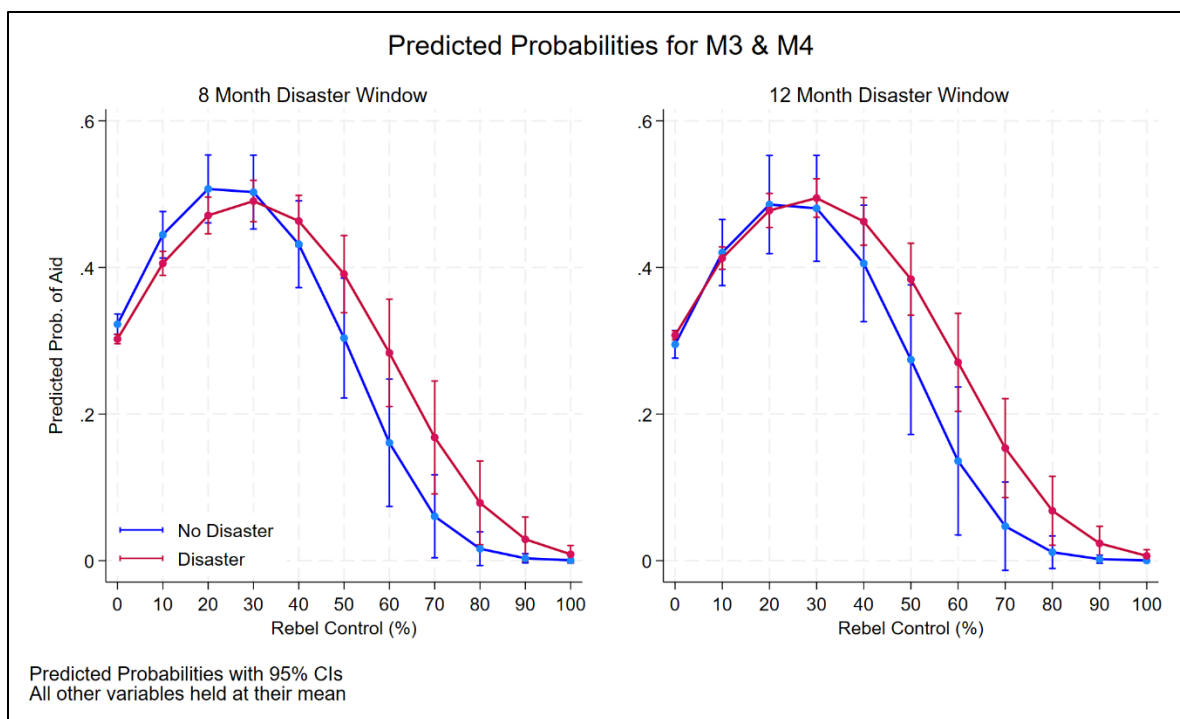


Figure A3: Predicted probability plots for the likelihood of a local aid project being installed, based on Models 3 and 4 of Table 3.

Supplementary Materials

Aid Disbursements as the Dependent Variable

Table B1 - Influence of Disasters & Control on Aid Disbursements per Month

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Disaster (1 Month)	16.441 (50.524)			
Disaster (4 Months)		67.352 ⁺ (37.260)		
Disaster (8 Months)			73.547 ⁺ (41.541)	
Disaster (12 Months)				30.227 (48.685)
Rebel Control (%)	85.438*** (14.140)	82.308*** (14.448)	75.707*** (15.235)	86.382*** (15.838)
Rebel Control ^{^2}	-1.383*** (0.301)	-1.309*** (0.305)	-1.192*** (0.316)	-1.388*** (0.325)
Disaster (1 Month) × Rebel Control	11.434 (15.659)			
Disaster (4 Months) × Rebel Control		18.962 ⁺ (11.093)		
Disaster (8 Months) × Rebel Control			23.814* (11.629)	
Disaster (12 Months) × Rebel Control				1.104 (12.845)
Disaster (1 Month) × Rebel Control ^{^2}	-0.318 (0.370)			
Disaster (4 Months) × Rebel Control ^{^2}		-0.515* (0.260)		
Disaster (8 Months) × Rebel Control ^{^2}			-0.563* (0.264)	
Disaster (12 Months) × Rebel Control ^{^2}				-0.069

				(0.292)
Avg. Distance to Capital	9.020***	9.140***	9.147***	9.042***
	(2.568)	(2.568)	(2.567)	(2.567)
Max. # of Excluded Ethnic Groups	150.863	161.180	163.715	154.630
	(1000.577)	(1000.477)	(1000.115)	(1000.208)
Avg. Maximum Nighttime Lights	60126.648***	60982.806***	61122.181***	60327.235***
	(7981.738)	(7990.619)	(7995.086)	(7993.498)
Avg. Forest Ground Cover	-12.345	-12.002	-11.908	-12.247
	(45.924)	(45.919)	(45.901)	(45.905)
Avg. Maximum Infant Mortality Rate	0.641	0.737	0.788	0.675
	(4.860)	(4.860)	(4.858)	(4.859)
Avg. Shrub Ground Cover	73.517	74.704	74.988	74.187
	(237.797)	(237.770)	(237.680)	(237.701)
Natural Resource Deposit	-2400.114**	-2423.071**	-2430.886**	-2406.931**
	(851.914)	(851.892)	(851.648)	(851.709)
Conflict-Related Deaths	100.825	98.734	100.150	101.237
	(89.746)	(89.763)	(89.745)	(89.757)
Constant	-3996.629 ⁺	-4174.291 ⁺	-4223.990 ⁺	-4051.155 ⁺
	(2403.242)	(2404.365)	(2404.978)	(2405.236)
Observations	29200	29200	29200	29200
R-Squared				

Standard errors in parentheses

⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

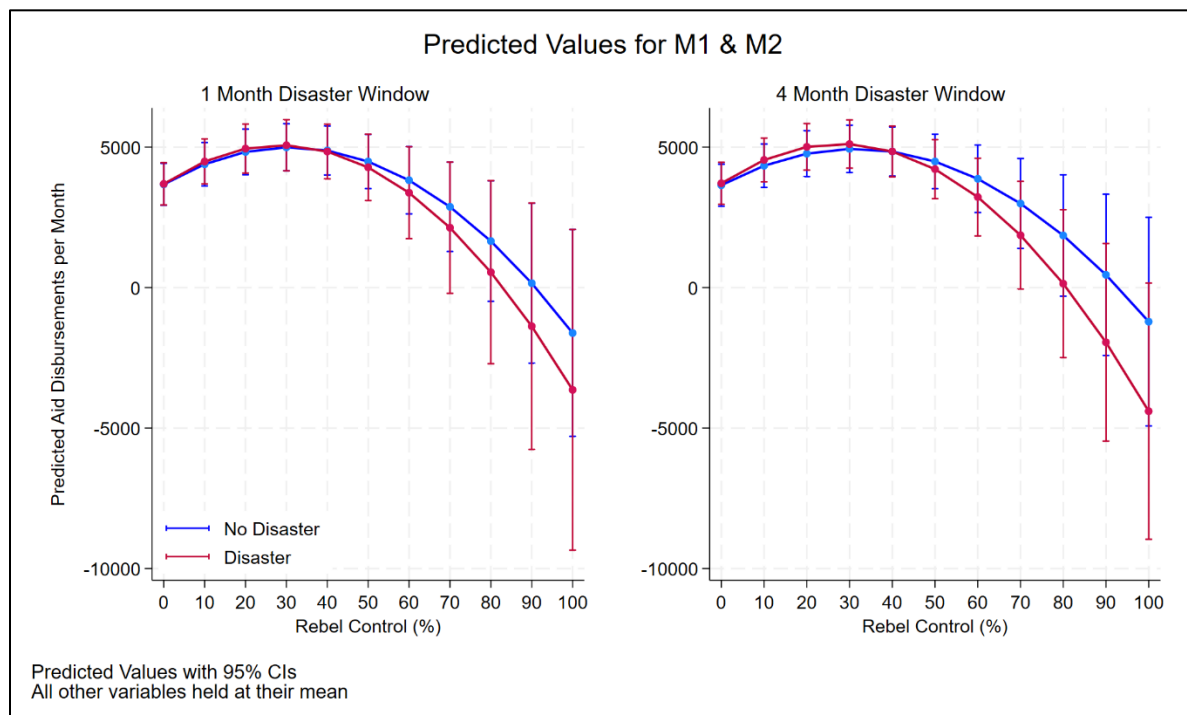


Figure B1: Predicted probability plots for the predicted amount of average aid disbursements per month, based on Models 1 and 2 of Table B1.

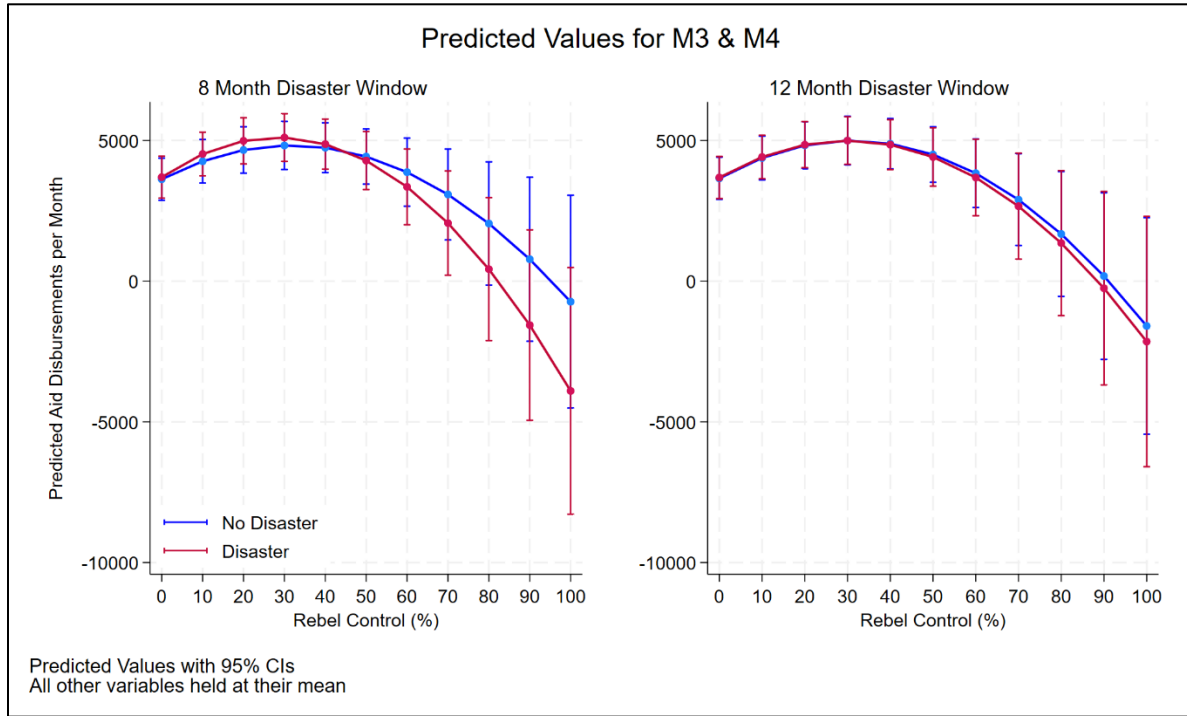


Figure B2: Predicted probability plots for the predicted amount of average aid disbursements per month, based on Models 3 and 4 of Table B1.

Table B2 - Influence of Disasters & Control on Aid Disbursements per Month (GDIS)

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
GDIS Disaster (1 Month)	-111.693* (44.467)			
GDIS Disaster (4 Months)		-237.041*** (40.430)		
GDIS Disaster (8 Months)			-211.304*** (55.138)	
GDIS Disaster (12 Months)				-242.155*** (73.481)
Rebel Control (%)	81.646*** (14.436)	76.464*** (16.378)	60.474** (18.516)	54.398* (21.502)
Rebel Control^2	-1.320*** (0.309)	-1.271*** (0.366)	-0.955* (0.413)	-0.855+ (0.459)
GDIS Disaster (1 Month) × Rebel Control	16.982 (13.221)			

GDIS Disaster (4 Months) × Rebel Control		14.587 (11.317)		
GDIS Disaster (8 Months) × Rebel Control			28.292* (13.302)	
GDIS Disaster (12 Months) × Rebel Control				32.713+ (17.153)
GDIS Disaster (1 Month) × Rebel Control^2	-0.326 (0.318)			
GDIS Disaster (4 Months) × Rebel Control^2		-0.193 (0.260)		
GDIS Disaster (8 Months) × Rebel Control^2			-0.492+ (0.297)	
GDIS Disaster (12 Months) × Rebel Control^2				-0.571 (0.361)
Avg. Distance to Capital	8.984*** (2.564)	9.083*** (2.566)	8.814*** (2.566)	8.922*** (2.567)
Max. # of Excluded Ethnic Groups	146.591 (999.475)	154.214 (999.901)	138.247 (999.698)	147.268 (1000.515)
Avg. Maximum Nighttime Lights	59861.682*** (7968.067)	60508.112*** (7972.054)	58871.644*** (7983.279)	59578.103*** (7978.051)
Avg. Forest Ground Cover	-12.496 (45.874)	-12.600 (45.894)	-12.668 (45.885)	-12.310 (45.922)
Avg. Maximum Infant Mortality Rate	0.570 (4.855)	0.483 (4.857)	0.411 (4.856)	0.494 (4.860)
Avg. Shrub Ground Cover	72.068 (237.538)	69.978 (237.640)	69.864 (237.592)	71.305 (237.786)
Natural Resource Deposit	-2393.757** (850.951)	-2406.778** (851.318)	-2374.988** (851.195)	-2390.893** (851.843)
Conflict-Related Deaths	99.227 (89.743)	96.373 (89.700)	100.031 (89.730)	102.564 (89.736)
Constant	-3900.459 (2400.083)	-3819.119 (2401.170)	-3529.503 (2404.445)	-3622.508 (2405.213)
Observations	29200	29200	29200	29200
R-Squared				

Standard errors in parentheses

+ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

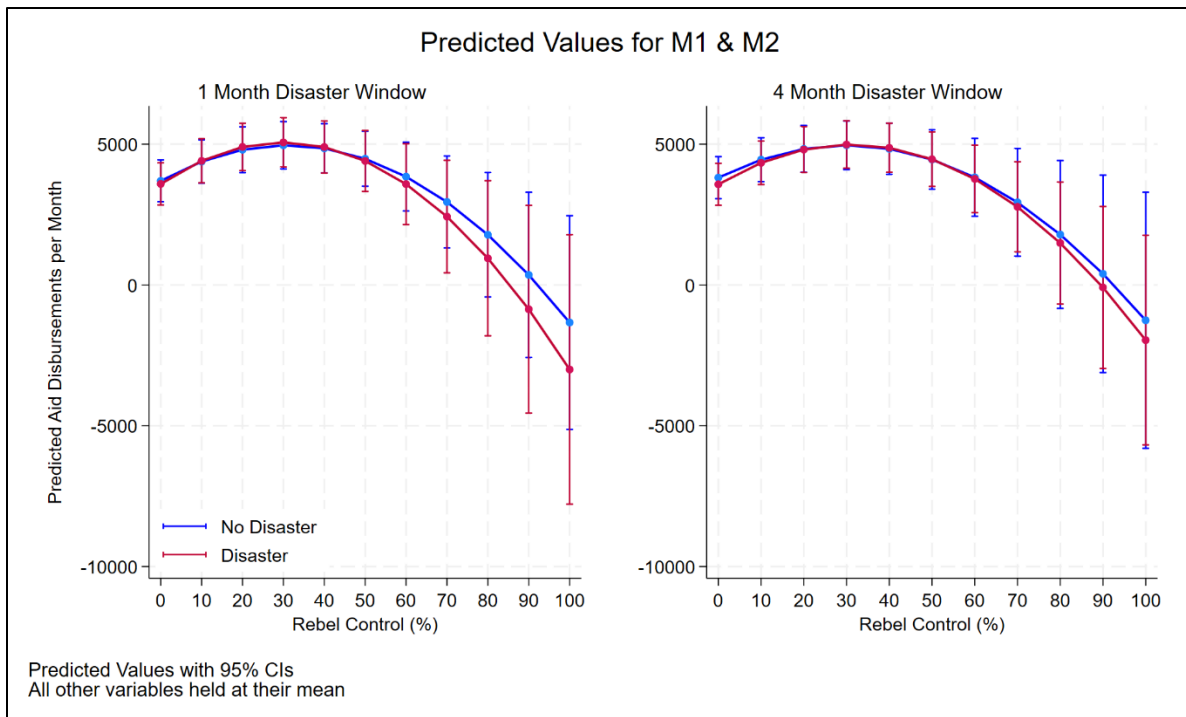


Figure B3: Predicted probability plots for the predicted amount of average aid disbursements per month, based on Models 1 and 2 of Table B2.

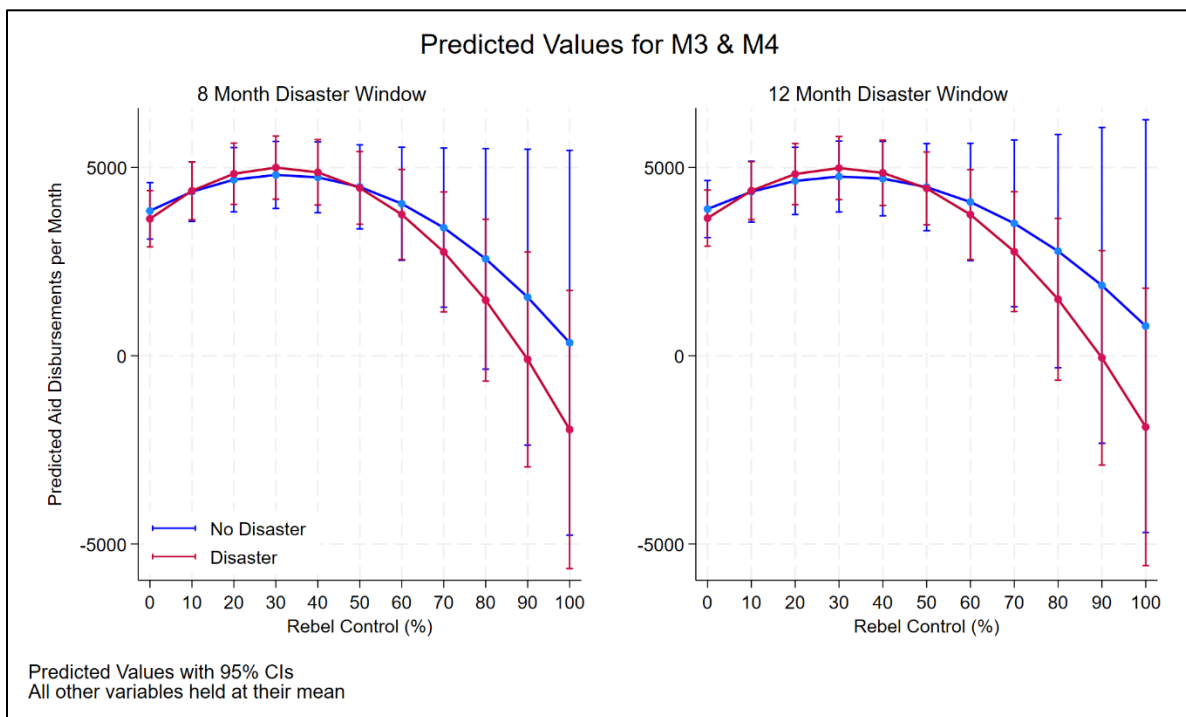


Figure B4: Predicted probability plots for the predicted amount of average aid disbursements per month, based on Models 3 and 4 of Table B2.

Disasters Measured via GDIS Coordinate Points

Table B3 - Influence of Disasters (GDIS - Coords) & Control on Presence of Aid Projects

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Presence of Project				
Disaster (1 Month)	0.217 (0.139)			
Disaster (4 Months)		0.068 ⁺ (0.041)		
Disaster (8 Months)			0.104 ^{**} (0.038)	
Disaster (12 Months)				0.186 ^{***} (0.055)
Rebel Control (%)	0.057 ^{***} (0.004)	0.059 ^{***} (0.004)	0.066 ^{***} (0.005)	0.072 ^{***} (0.006)
Rebel Control ^{^2}	-0.001 ^{***} (0.000)	-0.001 ^{***} (0.000)	-0.001 ^{***} (0.000)	-0.001 ^{***} (0.000)
Disaster (1 Month) × Rebel Control	0.076 (0.054)			
Disaster (4 Months) × Rebel Control		-0.010 (0.011)		
Disaster (8 Months) × Rebel Control			-0.021 ^{**} (0.008)	
Disaster (12 Months) × Rebel Control				-0.027 ^{***} (0.008)
Disaster (1 Month) × Rebel Control ^{^2}	-0.001 (0.001)			
Disaster (4 Months) × Rebel Control ^{^2}		0.000 (0.000)		
Disaster (8 Months) × Rebel Control ^{^2}			0.001 ^{**} (0.000)	
Disaster (12 Months) × Rebel Control ^{^2}				0.001 ^{***} (0.000)
Avg. Distance to Capital	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)

Max. # of Excluded Ethnic Groups	-0.196*** (0.033)	-0.197*** (0.033)	-0.198*** (0.033)	-0.198*** (0.033)
Avg. Maximum Nighttime Lights	0.706* (0.337)	0.709* (0.337)	0.714* (0.337)	0.719* (0.337)
Avg. Forest Ground Cover	0.014*** (0.002)	0.014*** (0.002)	0.014*** (0.002)	0.014*** (0.002)
Avg. Maximum Infant Mortality Rate	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Avg. Shrub Ground Cover	-0.171*** (0.009)	-0.171*** (0.009)	-0.171*** (0.009)	-0.171*** (0.009)
Natural Resource Deposit	-0.224*** (0.029)	-0.224*** (0.029)	-0.225*** (0.029)	-0.225*** (0.029)
Conflict-Related Deaths	0.197** (0.071)	0.197** (0.071)	0.196** (0.071)	0.193** (0.071)
Constant	-0.579*** (0.088)	-0.590*** (0.088)	-0.636*** (0.092)	-0.734*** (0.101)
Observations	29200	29200	29200	29200
Pseudo R-Squared	0.03	0.03	0.03	0.03

Standard errors in parentheses

⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

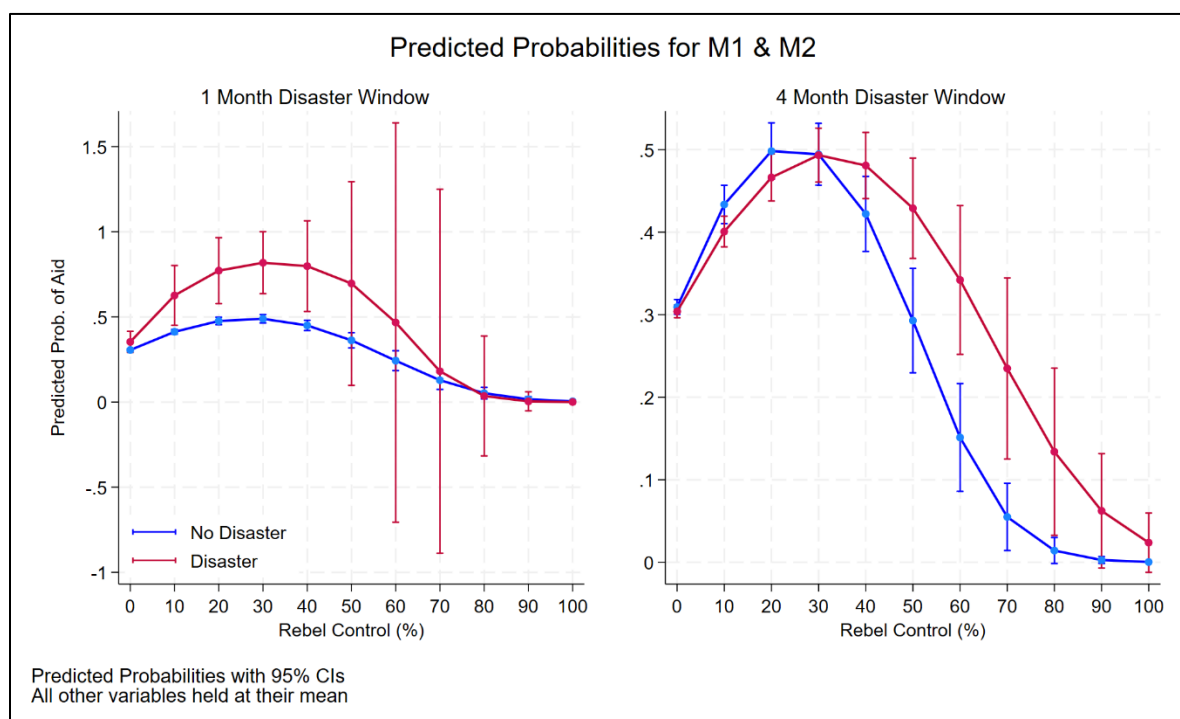


Figure B5: Predicted probability plots for the likelihood of local aid project presence, based on Models 1 and 2 of Table B3.

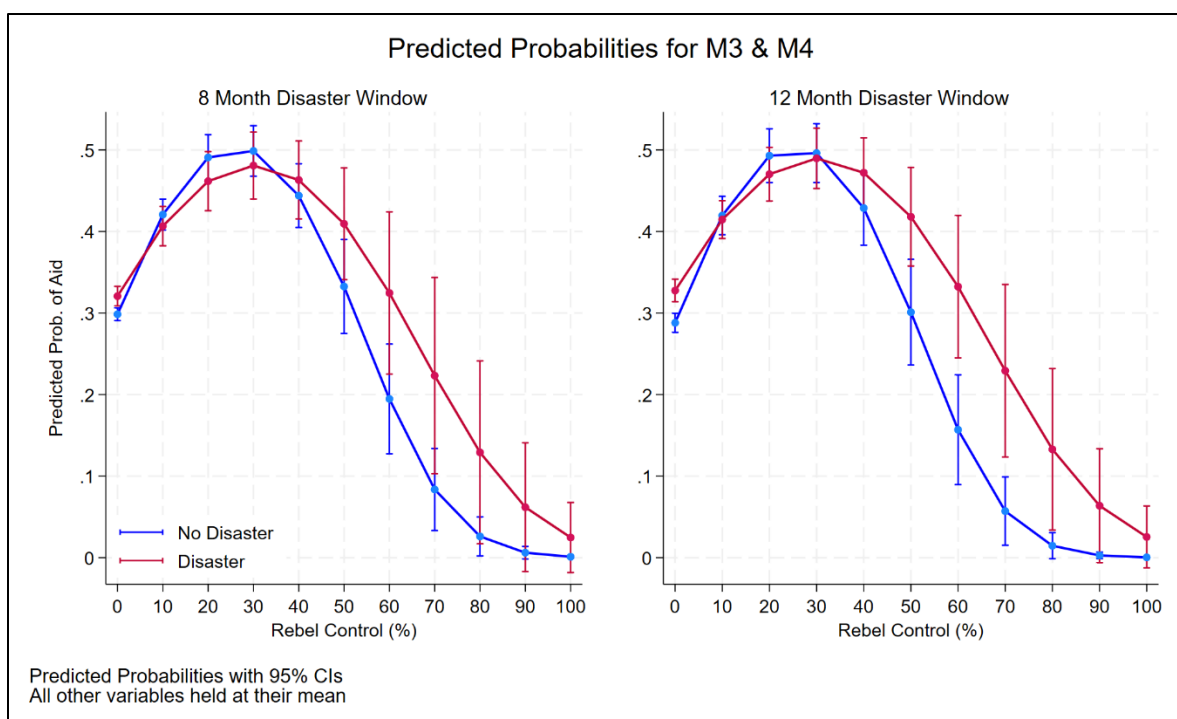


Figure B6: Predicted probability plots for the likelihood of local aid project presence, based on Models 3 and 4 of Table B3.

Substituting Time Cubic Polynomials for Year-level Fixed Effects

Table B4 - Influence of Disasters & Control on Presence of Aid Projects (Time Cubic Polynomials)

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Presence of Project				
Disaster (1 Month)	0.050 (0.040)			
Disaster (4 Months)		0.014 (0.028)		
Disaster (8 Months)			-0.040 (0.031)	
Disaster (12 Months)				-0.035 (0.036)
Rebel Control (%)	0.057*** (0.004)	0.054*** (0.005)	0.050*** (0.006)	0.040*** (0.007)
Rebel Control ²	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Disaster (1 Month) × Rebel Control	0.010 (0.013)			
Disaster (4 Months) × Rebel Control		0.012		

		(0.008)		
Disaster (8 Months) × Rebel Control			0.015 ⁺ (0.008)	
Disaster (12 Months) × Rebel Control				0.028 ^{**} (0.009)
Disaster (1 Month) × Rebel Control ^{^2}	-0.000 (0.000)			
Disaster (4 Months) × Rebel Control ^{^2}		-0.000 [*] (0.000)		
Disaster (8 Months) × Rebel Control ^{^2}			-0.001 ^{**} (0.000)	
Disaster (12 Months) × Rebel Control ^{^2}				-0.001 ^{***} (0.000)
Avg. Distance to Capital	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Max. # of Excluded Ethnic Groups	-0.197 ^{***} (0.033)	-0.198 ^{***} (0.033)	-0.202 ^{***} (0.033)	-0.203 ^{***} (0.033)
Avg. Maximum Nighttime Lights	0.678 [*] (0.338)	0.687 [*] (0.339)	0.758 [*] (0.340)	0.732 [*] (0.339)
Avg. Forest Ground Cover	0.014 ^{***} (0.002)	0.014 ^{***} (0.002)	0.014 ^{***} (0.002)	0.014 ^{***} (0.002)
Avg. Maximum Infant Mortality Rate	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Avg. Shrub Ground Cover	-0.171 ^{***} (0.009)	-0.171 ^{***} (0.009)	-0.171 ^{***} (0.009)	-0.172 ^{***} (0.009)
Natural Resource Deposit	-0.224 ^{***} (0.029)	-0.224 ^{***} (0.029)	-0.221 ^{***} (0.029)	-0.223 ^{***} (0.029)
Conflict-Related Deaths	0.196 ^{**} (0.071)	0.197 ^{**} (0.071)	0.197 ^{**} (0.071)	0.195 ^{**} (0.071)
Time	-0.005 (0.020)	-0.008 (0.020)	-0.007 (0.020)	-0.008 (0.020)
Time ^{^2}	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)
Time ^{^3}	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Constant	-0.564 ^{***} (0.105)	-0.554 ^{***} (0.105)	-0.541 ^{***} (0.105)	-0.534 ^{***} (0.105)
Observations	29200	29200	29200	29200
Pseudo R-Squared	0.03	0.03	0.03	0.03

Standard errors in parentheses

⁺ $p < .10$, ^{*} $p < .05$, ^{**} $p < .01$, ^{***} $p < .001$

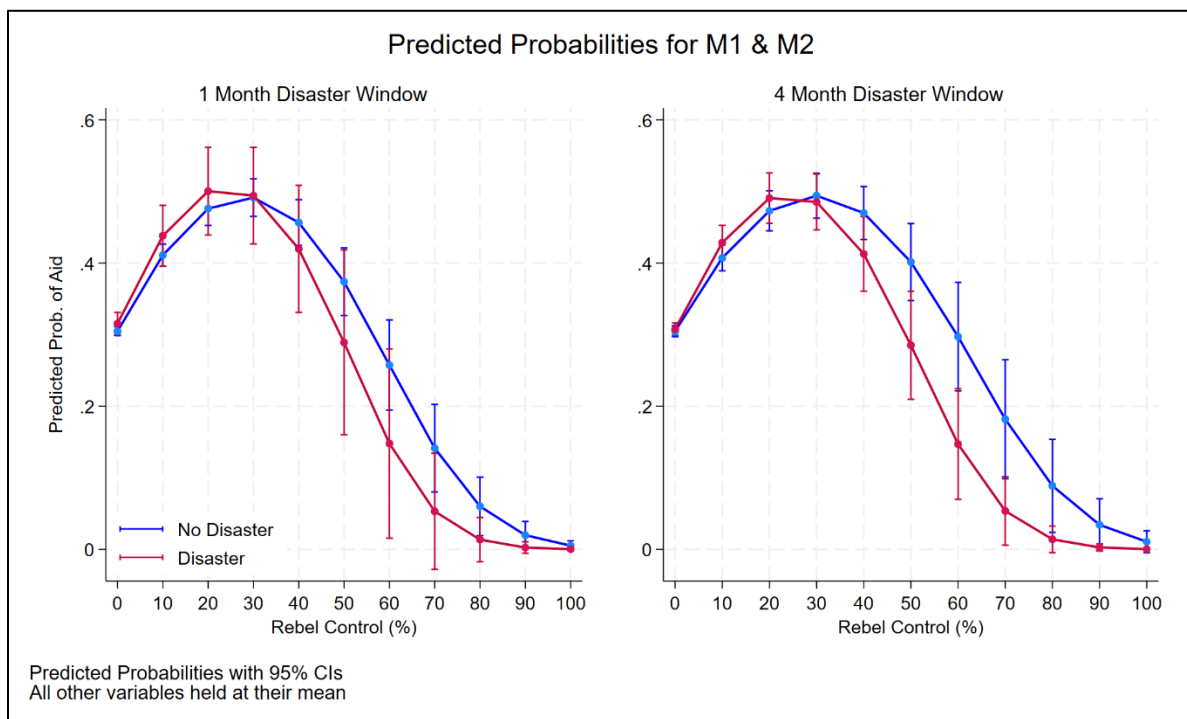


Figure B7: Predicted probability plots for the likelihood of local aid project presence, based on Models 1 and 2 of Table B4.

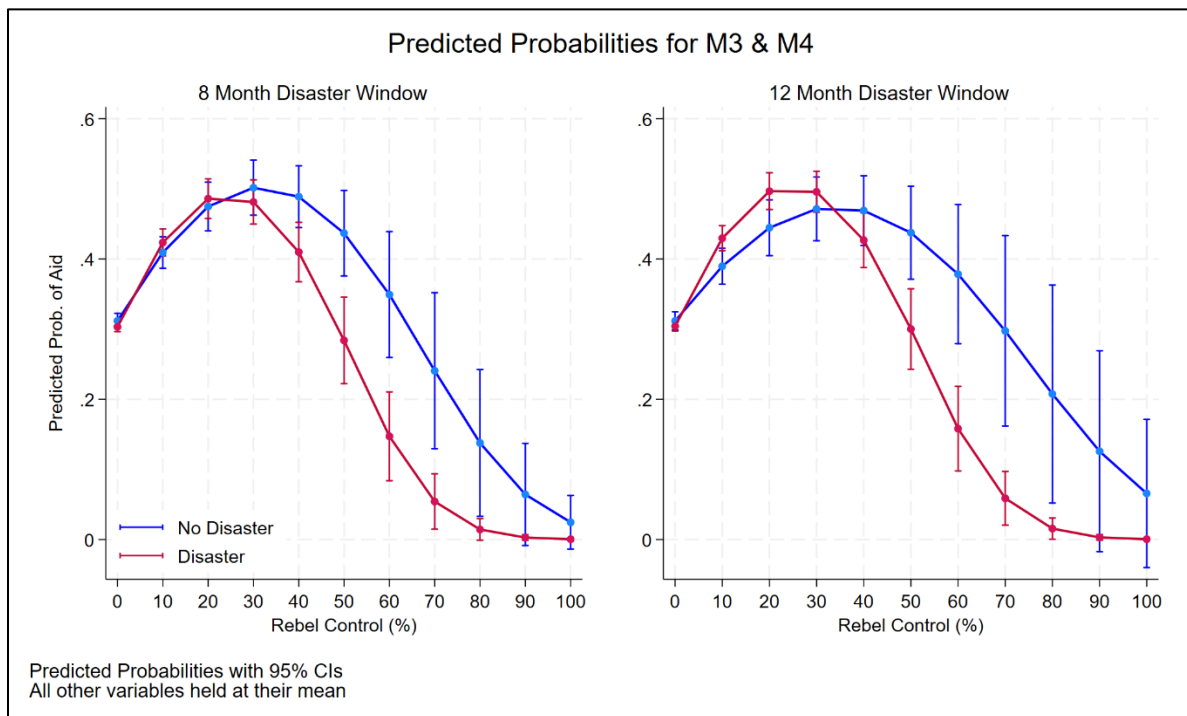


Figure B8: Predicted probability plots for the likelihood of local aid project presence, based on Models 3 and 4 of Table B4.

Inclusion of Region-level Fixed Effects

Table B5 - Influence of Disasters & Control on Presence of Aid Projects (Region F.E.)

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Presence of Project				
Disaster (1 Month)	0.001 (0.041)			
Disaster (4 Months)		-0.018 (0.030)		
Disaster (8 Months)			-0.083** (0.032)	
Disaster (12 Months)				-0.105** (0.037)
Rebel Control (%)	0.026*** (0.005)	0.022*** (0.005)	0.020** (0.006)	0.017* (0.007)
Rebel Control^2	-0.001*** (0.000)	-0.000*** (0.000)	-0.000** (0.000)	-0.000+ (0.000)
Disaster (1 Month) × Rebel Control	0.009 (0.013)			
Disaster (4 Months) × Rebel Control		0.015+ (0.009)		
Disaster (8 Months) × Rebel Control			0.016+ (0.009)	
Disaster (12 Months) × Rebel Control				0.018+ (0.009)
Disaster (1 Month) × Rebel Control^2	-0.000 (0.000)			
Disaster (4 Months) × Rebel Control^2		-0.001* (0.000)		
Disaster (8 Months) × Rebel Control^2			-0.001** (0.000)	
Disaster (12 Months) × Rebel Control^2				-0.001** (0.000)
Avg. Distance to Capital	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Max. # of Excluded Ethnic Groups	-0.393*** (0.042)	-0.397*** (0.042)	-0.402*** (0.042)	-0.405*** (0.042)
Avg. Maximum Nighttime Lights	6.727*** (0.459)	6.738*** (0.460)	6.799*** (0.460)	6.786*** (0.460)

Avg. Forest Ground Cover	0.006*** (0.002)	0.006*** (0.002)	0.006*** (0.002)	0.006*** (0.002)
Avg. Maximum Infant Mortality Rate	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
Avg. Shrub Ground Cover	-0.076*** (0.011)	-0.076*** (0.011)	-0.075*** (0.011)	-0.076*** (0.011)
Natural Resource Deposit	-0.138*** (0.034)	-0.138*** (0.034)	-0.135*** (0.034)	-0.135*** (0.034)
Conflict-Related Deaths	0.167* (0.071)	0.168* (0.071)	0.168* (0.070)	0.167* (0.070)
Constant	-3.157*** (0.195)	-3.150*** (0.196)	-3.108*** (0.196)	-3.098*** (0.197)
Observations	29200	29200	29200	29200
Pseudo R-Squared	0.10	0.10	0.10	0.10

Standard errors in parentheses

+ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

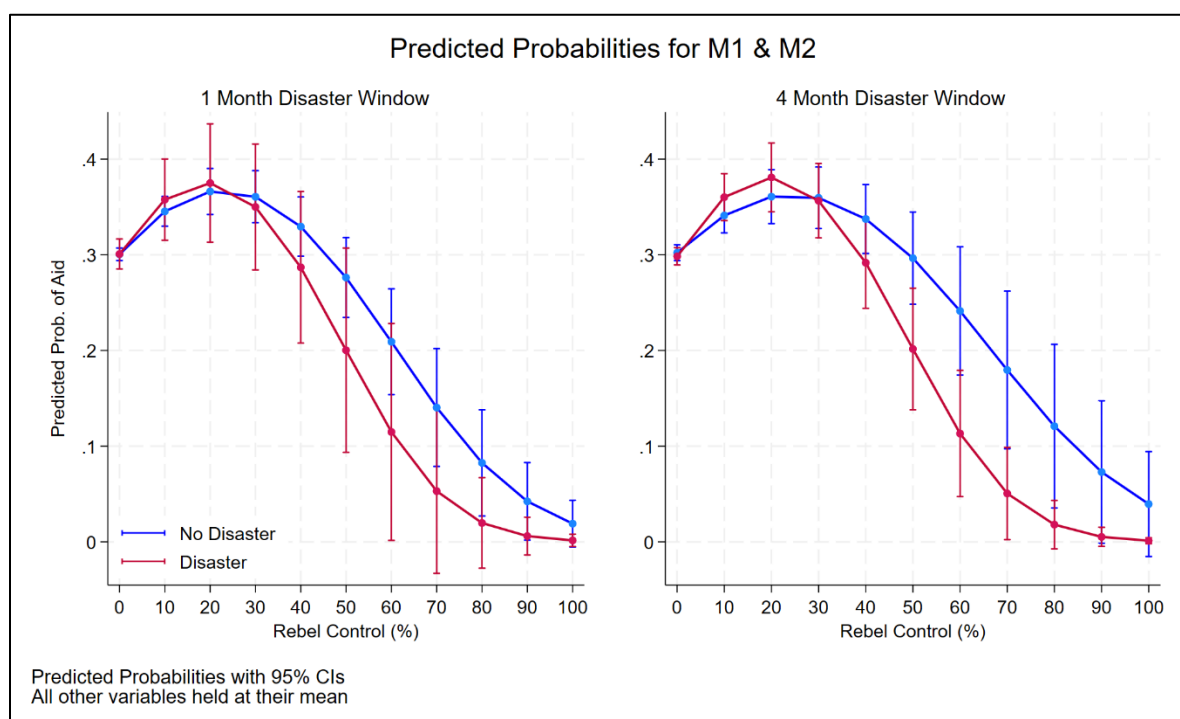


Figure B9: Predicted probability plots for the likelihood of local aid project presence, based on Models 1 and 2 of Table B5.

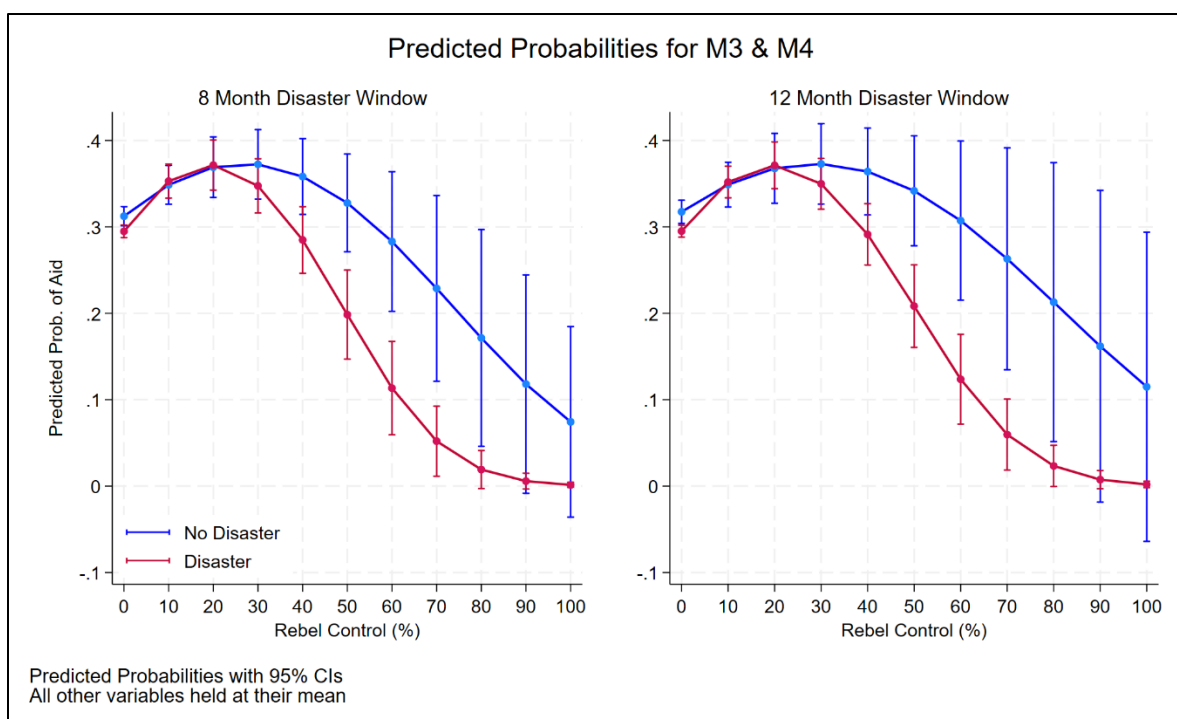


Figure B10: Predicted probability plots for the likelihood of local aid project presence, based on Models 3 and 4 of Table B5.

Municipality-Year Analysis

Table B6 - Influence of Disasters & Control on Presence of Aid Projects
(Municipality-Year - Disasters Lagged One Year)

	(1) M1: IMERG Disasters	(2) M2: GDIS Disasters
Presence of Project		
Disaster in the Previous Year (IMERG)	-0.165 (0.142)	
Disaster in the Previous Year (GDIS)		-0.212 (0.151)
Rebel Control (%)	0.079* (0.035)	-0.042 (0.064)
Rebel Control ²	-0.001 (0.001)	-0.000 (0.002)
Disaster in the Previous Year (IMERG) × Rebel Control	-0.015 (0.039)	
Disaster in the Previous Year (GDIS) × Rebel Control		0.118+ (0.066)

Disaster in the Previous Year (IMERG) $\times$ Rebel Control ²	0.000 (0.001)	
Disaster in the Previous Year (GDIS) $\times$ Rebel Control ²		-0.001 (0.002)
Avg. Distance to Capital	-0.000 (0.000)	0.000 (0.000)
Max. # of Excluded Ethnic Groups	-0.221 ⁺ (0.119)	-0.239* (0.120)
Avg. Maximum Nighttime Lights	0.527 (1.235)	0.669 (1.241)
Avg. Forest Ground Cover	0.014** (0.005)	0.015** (0.005)
Avg. Maximum Infant Mortality Rate	-0.000 (0.001)	-0.000 (0.001)
Avg. Shrub Ground Cover	-0.198*** (0.034)	-0.205*** (0.034)
Natural Resource Deposit	-0.198 ⁺ (0.105)	-0.218* (0.105)
Conflict-Related Deaths	0.196** (0.076)	0.202** (0.075)
Constant	-0.372 (0.326)	-0.320 (0.362)
Observations	2336	2336
Pseudo R-Squared	0.04	0.04

Standard errors in parentheses

⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

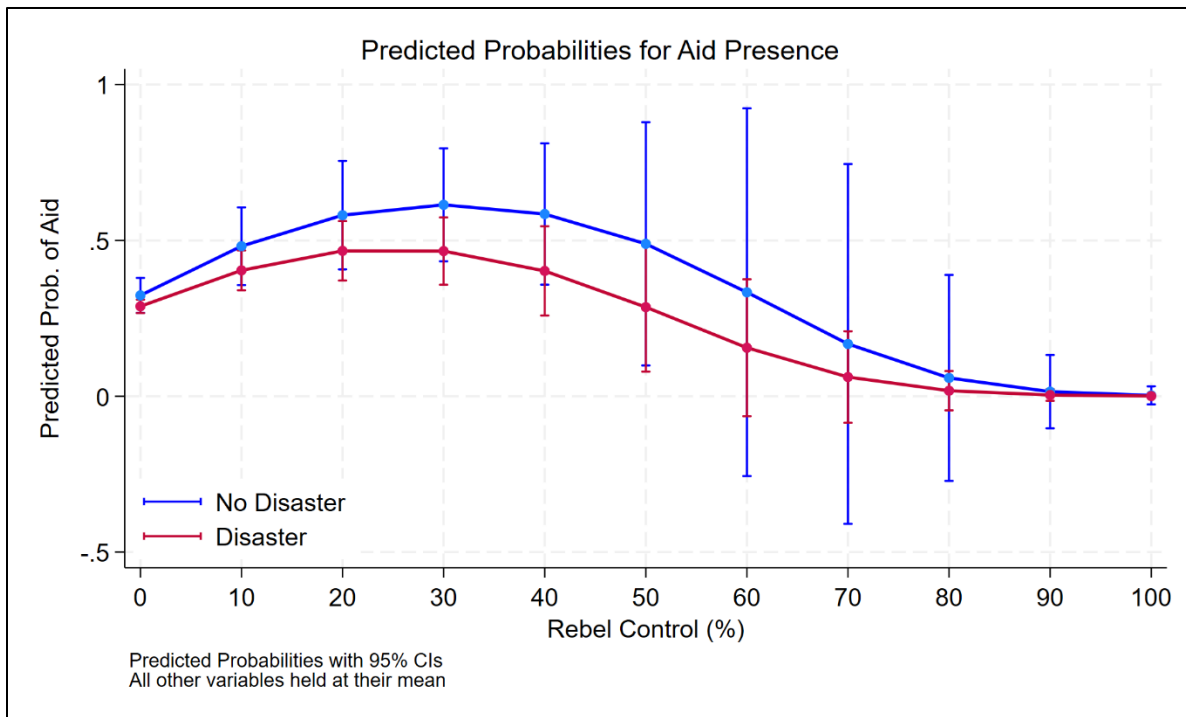


Figure B9: Predicted probability plots for the likelihood of local aid project presence, based on Model 1 of Table B6.

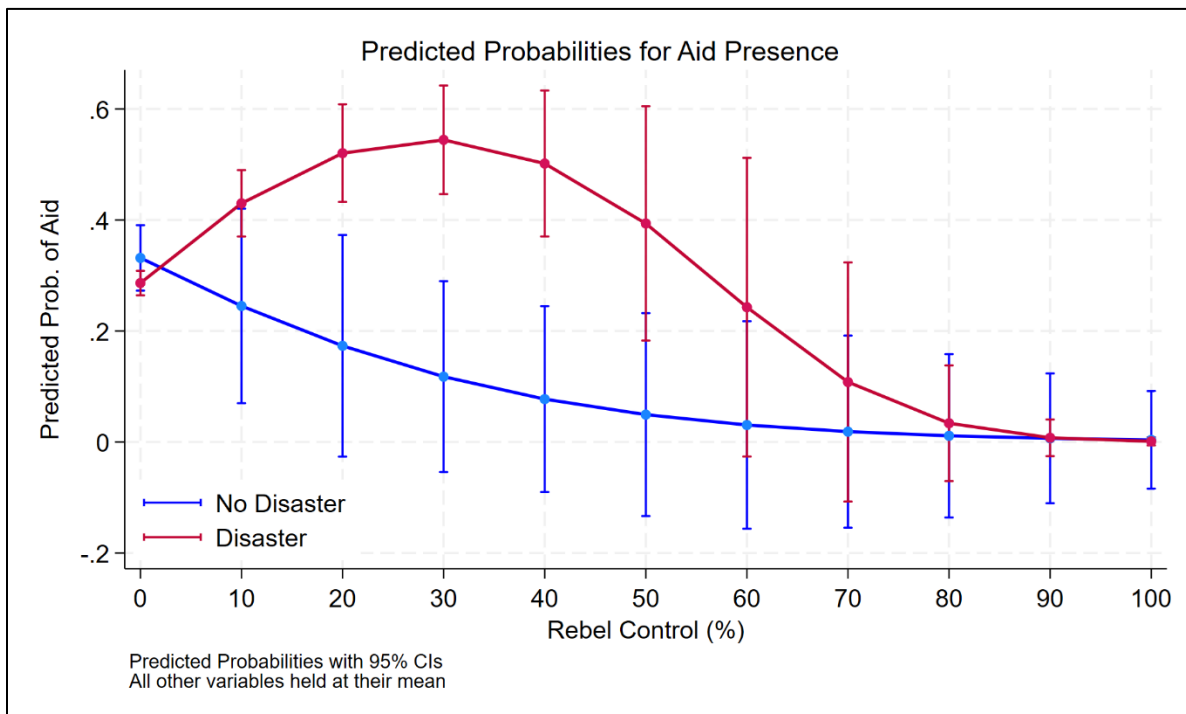


Figure B10: Predicted probability plots for the likelihood of local aid project presence, based on Model 2 of Table B6.

Table B7 - Influence of Disasters & Control on Presence of Aid Projects
(Municipality-Year - Disasters and Control Lagged One Year)

	(1) M1: IMERG Disasters	(2) M2: GDIS Disasters
Presence of Project		
Disaster in the Previous Year (IMERG)	-0.193 (0.142)	
Disaster in the Previous Year (GDIS)		-0.185 (0.151)
Rebel Control in the Previous Year (%)	0.056* (0.027)	0.011 (0.051)
Rebel Control ²	-0.001 (0.001)	-0.001 (0.001)
Disaster in the Previous Year (IMERG) × Rebel Control	0.003 (0.032)	
Disaster in the Previous Year (GDIS) × Rebel Control		0.049 (0.053)
Disaster in the Previous Year (IMERG) × Rebel Control ²	-0.000 (0.001)	
Disaster in the Previous Year (GDIS) × Rebel Control ²		0.000 (0.001)
Avg. Distance to Capital	0.000 (0.000)	0.000 (0.000)
Max. # of Excluded Ethnic Groups	-0.223 ⁺ (0.119)	-0.235* (0.120)
Avg. Maximum Nighttime Lights	0.509 (1.236)	0.606 (1.243)
Avg. Forest Ground Cover	0.015** (0.005)	0.016** (0.005)
Avg. Maximum Infant Mortality Rate	-0.000 (0.001)	-0.000 (0.001)
Avg. Shrub Ground Cover	-0.198*** (0.034)	-0.203*** (0.034)
Natural Resource Deposit	-0.200 ⁺ (0.105)	-0.220* (0.105)
Conflict-Related Deaths	0.204** (0.076)	0.213** (0.076)
Constant	-0.358	-0.353

	(0.328)	(0.362)
Observations	2336	2336
Pseudo R-Squared	0.04	0.04

Standard errors in parentheses

⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

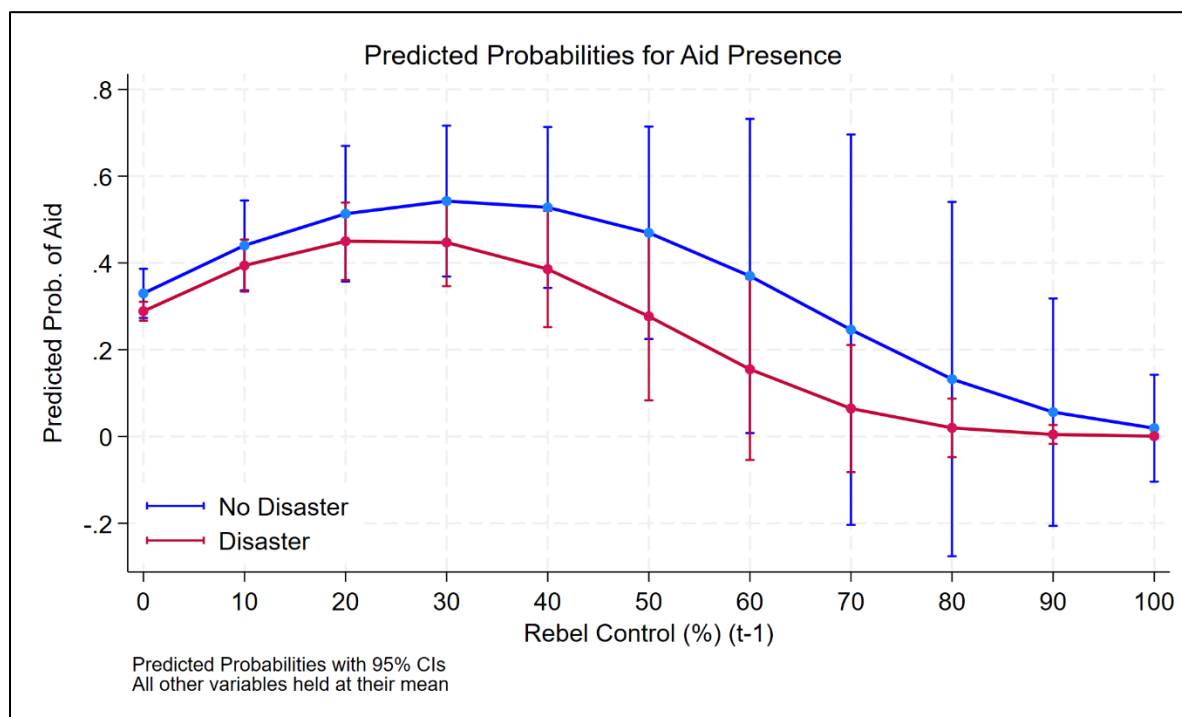


Figure B11: Predicted probability plots for the likelihood of local aid project presence, based on Model 1 of Table B7.

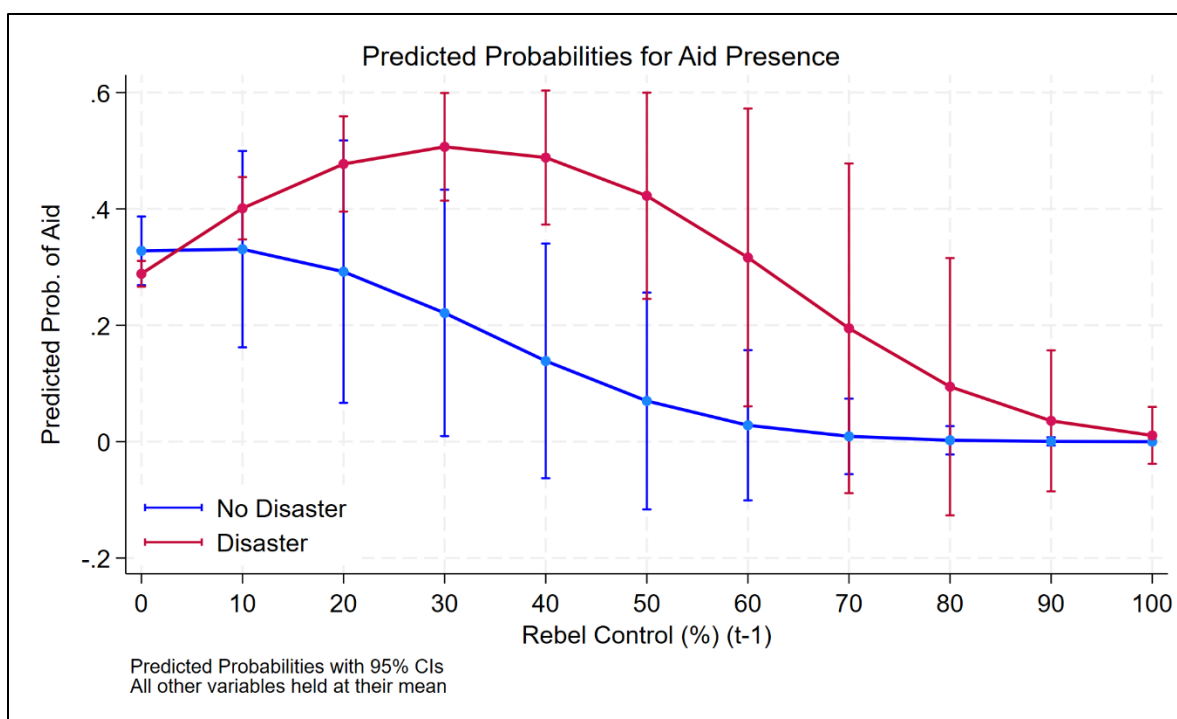


Figure B12: Predicted probability plots for the likelihood of local aid project presence, based on Model 2 of Table B7.

Standard Errors Clustered by Municipality

Table B8 - Influence of Disasters & Control on Presence of Aid Projects (Clustered by Municipality)

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Presence of Project				
Disaster (1 Month)	0.052* (0.022)			
Disaster (4 Months)		0.012 (0.033)		
Disaster (8 Months)			-0.040 (0.051)	
Disaster (12 Months)				-0.036 (0.071)
Rebel Control (%)	0.057** (0.019)	0.054** (0.020)	0.050* (0.022)	0.040+ (0.023)
Rebel Control^2	-0.001* (0.000)	-0.001+ (0.000)	-0.001 (0.001)	-0.001 (0.001)
Disaster (1 Month) × Rebel Control	0.010 (0.012)			
Disaster (4 Months) × Rebel Control		0.012		

		(0.015)		
Disaster (8 Months) × Rebel Control			0.016 (0.020)	
Disaster (12 Months) × Rebel Control				0.028 (0.022)
Disaster (1 Month) × Rebel Control ²	-0.000 (0.000)			
Disaster (4 Months) × Rebel Control ²		-0.000 (0.000)		
Disaster (8 Months) × Rebel Control ²			-0.001 (0.001)	
Disaster (12 Months) × Rebel Control ²				-0.001 (0.001)
Avg. Distance to Capital	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Max. # of Excluded Ethnic Groups	-0.197 (0.163)	-0.198 (0.163)	-0.202 (0.163)	-0.203 (0.163)
Avg. Maximum Nighttime Lights	0.676 (1.662)	0.689 (1.661)	0.758 (1.660)	0.733 (1.661)
Avg. Forest Ground Cover	0.014 ⁺ (0.008)	0.014 ⁺ (0.008)	0.014 ⁺ (0.008)	0.014 ⁺ (0.008)
Avg. Maximum Infant Mortality Rate	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
Avg. Shrub Ground Cover	-0.171*** (0.045)	-0.171*** (0.045)	-0.171*** (0.045)	-0.172*** (0.045)
Natural Resource Deposit	-0.224 (0.143)	-0.224 (0.143)	-0.221 (0.143)	-0.223 (0.143)
Conflict-Related Deaths	0.197* (0.082)	0.198* (0.083)	0.198* (0.083)	0.196* (0.082)
Constant	-0.574 (0.429)	-0.573 (0.429)	-0.550 (0.429)	-0.548 (0.429)
Observations	29200	29200	29200	29200
Pseudo R-Squared	0.03	0.03	0.03	0.03

Standard errors in parentheses

Standard errors clustered by District

⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

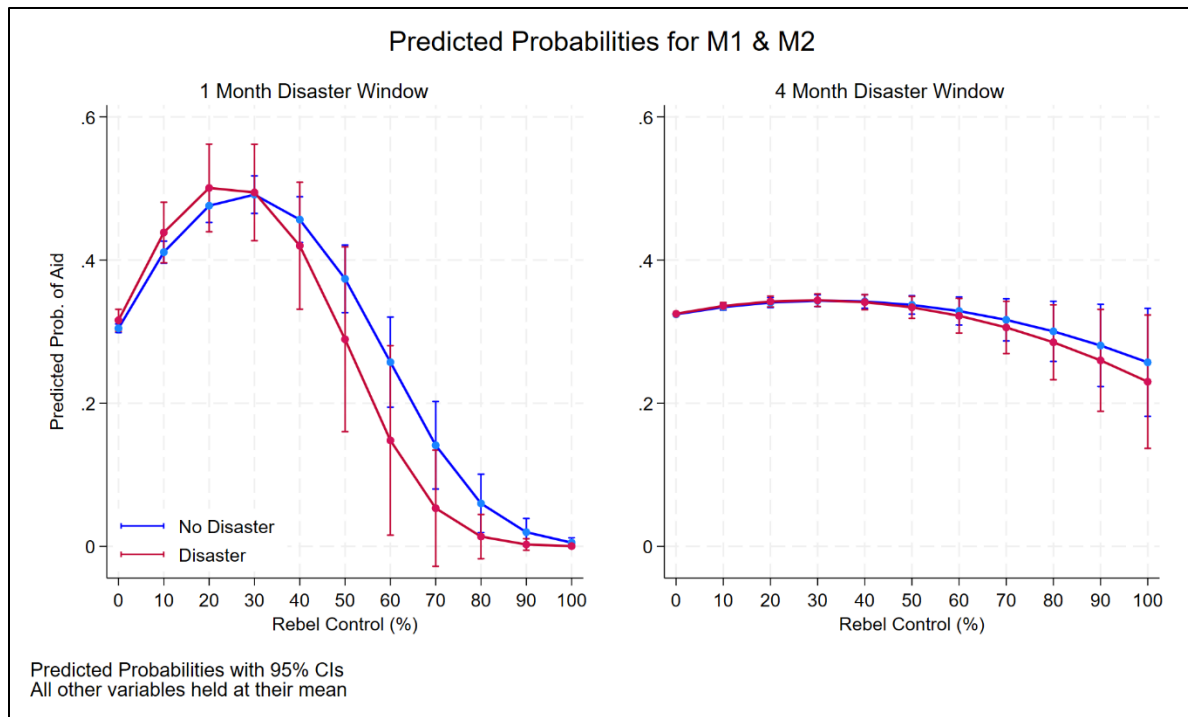


Figure B13: Predicted probability plots for the likelihood of local aid project presence, based on Models 1 and 2 of Table B8.

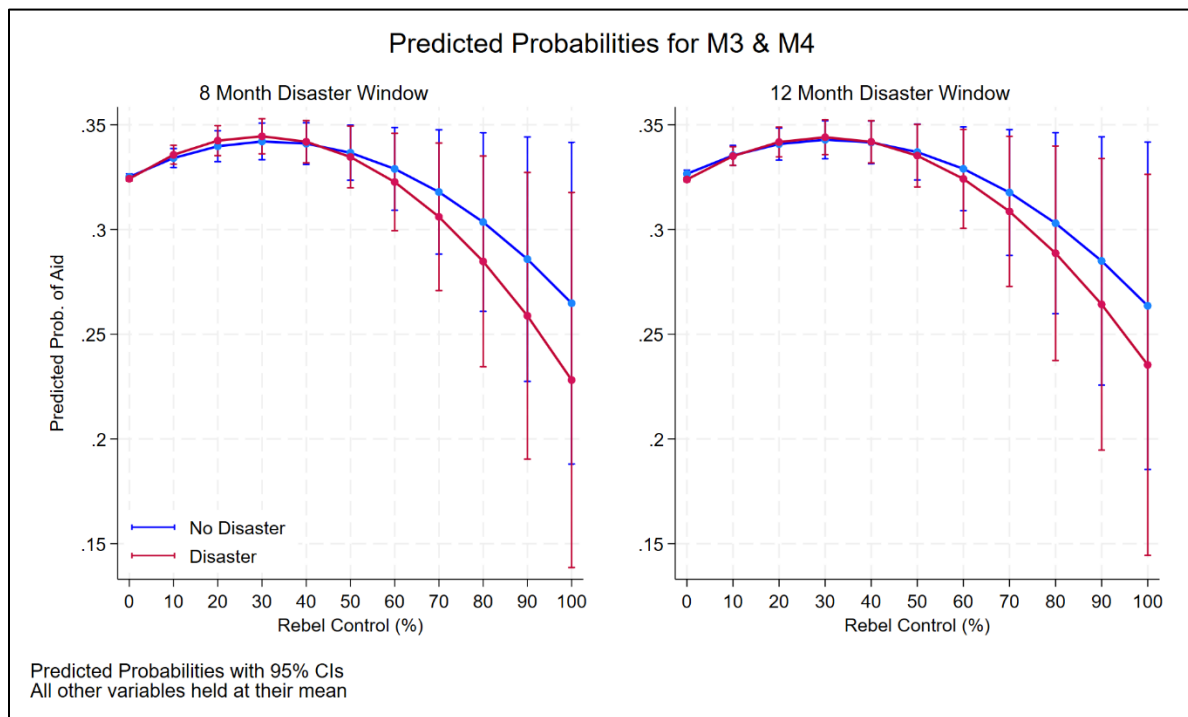


Figure B14: Predicted probability plots for the likelihood of local aid project presence, based on Models 3 and 4 of Table B8.

Aid Disbursements as the Dependent Variable (TWFE Models)

Table B9 - Influence of Disasters & Control on Aid Disbursements (TWFE)

	(1) M1: 1 Month Window	(2) M2: 4 Months	(3) M3: 8 Months	(4) M4: 12 Months
Disaster (1 Month)	26.626 (50.617)			
Disaster (4 Months)		79.090* (37.417)		
Disaster (8 Months)			89.304* (41.782)	
Disaster (12 Months)				46.985 (48.947)
Rebel Control (%)	80.712*** (14.257)	77.692*** (14.557)	71.393*** (15.333)	82.738*** (15.922)
Rebel Control^2	-1.279*** (0.303)	-1.205*** (0.307)	-1.094*** (0.318)	-1.301*** (0.326)
Disaster (1 Month) × Rebel Control	10.698 (15.659)			
Disaster (4 Months) × Rebel Control		17.924 (11.095)		
Disaster (8 Months) × Rebel Control			22.243+ (11.635)	
Disaster (12 Months) × Rebel Control				-1.225 (12.859)
Disaster (1 Month) × Rebel Control^2	-0.302 (0.370)			
Disaster (4 Months) × Rebel Control^2		-0.492+ (0.260)		
Disaster (8 Months) × Rebel Control^2			-0.527* (0.265)	
Disaster (12 Months) × Rebel Control^2				-0.022 (0.292)
Avg. Maximum Nighttime Lights	93322.202*** (12960.503)	95975.416*** (13005.557)	96644.285*** (13037.237)	94296.978*** (13026.773)
Conflict-Related Deaths	97.929 (89.742)	95.523 (89.757)	97.068 (89.738)	98.297 (89.751)
Constant	-5252.593*** (1239.776)	-5538.926*** (1245.527)	-5628.415*** (1250.591)	-5377.357*** (1250.735)
Observations	29200	29200	29200	29200

Pseudo R-Squared

Standard errors in parentheses

⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

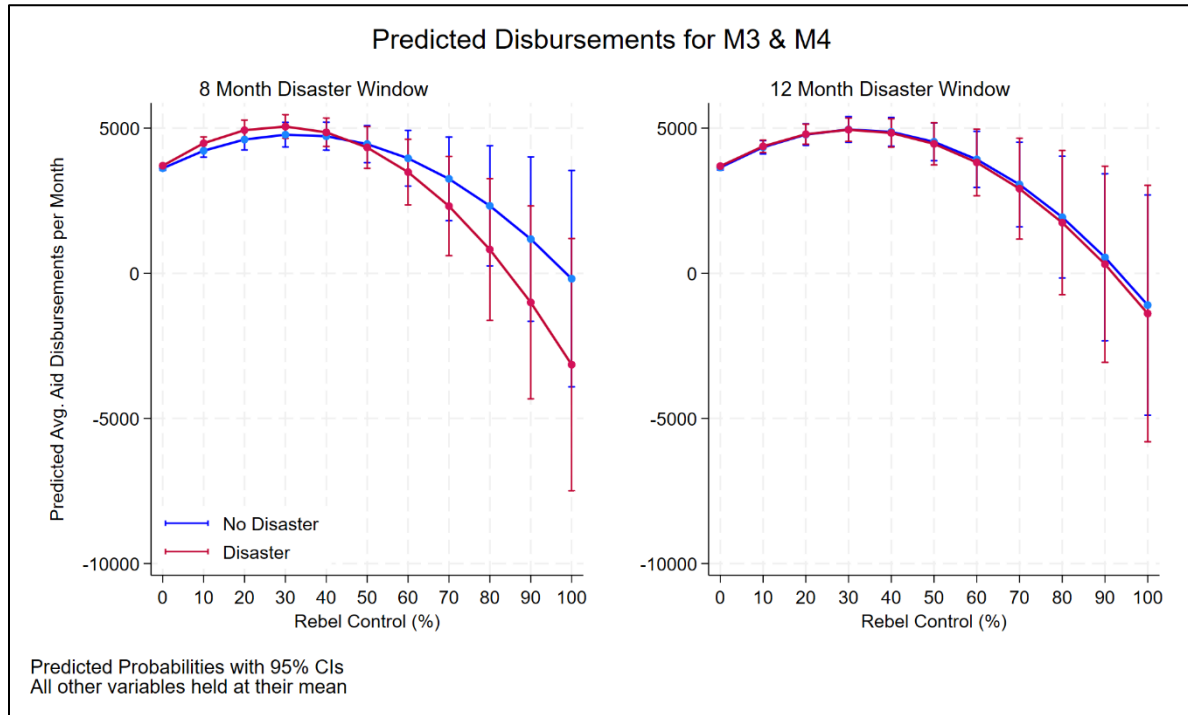


Figure B15: Predicted probability plots for the likelihood of local aid project presence, based on Models 1 and 2 of Table B9.

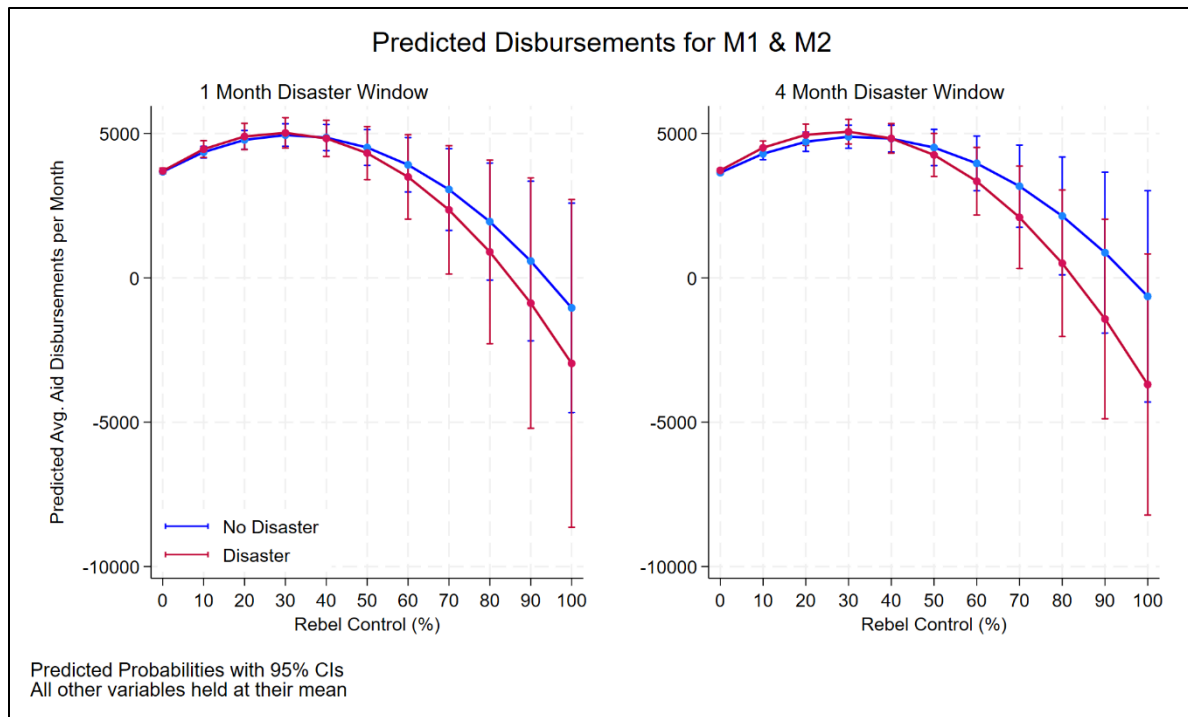


Figure B16: Predicted probability plots for the likelihood of local aid project presence, based on Models 3 and 4 of Table B9.